Robust Offline Policy Learning with Multisite Data

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Abstract

Robust offline policy learning (OPL) aims to learn policies from historical logged data that remain reliable under shifts between training and deployment environments. Existing robust OPL methods primarily study single-source adaptation or generalization, addressing challenges such as covariate shift, concept drift, or joint distribution shifts from one source domain to a target domain. This paper studies a more nuanced multi-source setting: labeled data are collected from several source domains, whereas the target domain provides only unlabeled covariates. Rather than relying on a single-source robustness criterion, we model the target outcome mechanism as an unknown convex combination of source-specific reward mechanisms. This leads to a minimax source-mixture objective that optimizes the worst-case policy value over the convex hull of source mechanisms. We derive a novel multi-source doubly robust estimator that efficiently integrates information across heterogeneous logging policies. Theoretically, we establish the asymptotic normality of our estimator and provide a regret upper bound for the learned policy. Experiments demonstrate that the proposed method outperforms the baselines and maintains robust performance across the simplex of unknown source weights.

1 Introduction

Offline policy learning (OPL) aims to learn an individualized treatment rule from historical logged data that maximizes expected reward under a target deployment population, with applications in precision medicine, online marketplaces, and public policy [Manski, 2004, Athey and Wager, 2021, Kitagawa and Tetenov, 2018, Kallus and Uehara, 2020, Dudík et al., 2011]. A central obstacle to the reliable deployment of the learned policy lies in the distribution shift: data are collected under one or multiple training environments, while the policy is executed in a new environment where the covariate distribution or outcome mechanism may differ. Such shifts can severely degrade the quality of decision-making.

To guard against unknown and potentially substantial discrepancies between training and deployment environments, recent work has focused on robust offline policy learning, which aims to learn policies that perform well under worst-case shifts from the source distribution to the target. Many existing robust OPL methods are single-source in nature: they model uncertainty as covariate shift [Guo et al., 2024], concept drift [Wang et al., 2025a], or more general joint distribution shifts [Si et al., 2023]. In many real-world applications, however, observational data come from multiple heterogeneous sites—for example, different hospitals, regions, platforms, or time periods. And the target outcome mechanism may not coincide with any single source. This motivates a multisite robust policy learning criterion that protects against unknown source-mixture ambiguity rather than conventional robustness around one empirical distribution.

In this paper, we propose a Robust Offline Policy Learning method tailored to multi-source settings with an unlabeled target domain. While target covariates are fully accessible—allowing us to explicitly measure and correct for covariate shift—the target conditional outcome mechanism remains

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fundamentally unobservable. Therefore, the core challenge we address is not covariate shift but this unobserved concept drift across heterogeneous domains. We tackle this by modeling the target outcome mechanism as an unknown convex combination of source mechanisms, effectively capturing environmental ambiguity while relaxing the restrictive assumption of strict transportability. This convex formulation is practical for modeling interpolation across environments; for example, a newly established target hospital's patient profile typically represents a mixture of established urban and rural sites rather than perfectly matching any single source. Leveraging this structural assumption, rather than simply aggregating site-specific predictions, we directly optimize the worst-case policy value over the convex hull of source domains. Specifically, we formulate a min-max objective over the simplex of unknown mixture weights to derive a policy that minimizes a theoretical regret upper bound.

Our contributions are summarized as follows:

Robust Estimation with multi-site data: we formulate multi-source offline policy learning with unlabeled target covariates under a convex-hull uncertainty set over source reward mechanisms.

Theoretical Guarantees: We construct a doubly robust estimator for the multiple-source value function, establish asymptotic normality guarantees, and derive a rigorous regret upper bound for the learned policy. We show that our method achieves a fast convergence rate of $O(\kappa(\Pi)/\sqrt{n_t})$ where $\kappa(\Pi)$ is a measure quantifying the policy class complexity, and n_t denotes the target sample size.

Empirical Validation: We demonstrate through experiments that our method outperforms the baselines. Our approach is particularly effective in conflicting signal scenarios where different sources may suggest divergent optimal actions, successfully improving worst-case and lower-tail performance under unknown source mixtures.

1.1 Related Literature

Offline Policy Learning. The primary objective of offline policy learning (OPL) is to extract welfare-maximizing decision rules from historical logged data. Seminal frameworks typically rely on inverse propensity weighting or doubly robust estimators to ensure unbiased evaluation and learning [Swaminathan and Joachims, 2015, Kitagawa and Tetenov, 2018, Athey and Wager, 2021]. However, these classical guarantees are largely predicated on the assumption that the training and deployment environments are identically distributed. Departing from this idealization, recent literature focuses on addressing distributional shift. Pioneering works by [Si et al., 2023] and Kallus et al. [2022] propose optimizing the policy against a worst-case distribution within a statistical uncertainty set centered around the training data. [Wang et al., 2025a] allows for concept drift (the conditional relationship between the outcome Y and the covariate X) from the source domain to the target domain. [Liu et al., 2025] established a doubly robust estimator for covariate shift while the conditional reward mechanism remains invariant. Despite their theoretical rigor, these approaches are inherently single-source centric: they model uncertainty as perturbations around a single dataset.

Multiple Source Adaptation. [Wang et al., 2025b] demonstrated that utilizing unlabeled target covariates can significantly enhance outcome prediction by aggregating source models through explained-variance robustness. Similarly, in the causal inference literature, [Zhang et al., 2024] introduced a minimax regret framework to generalize conditional average treatment effects (CATE) across multisite data. In the context of offline policy evaluation (OPE), [Kallus et al., 2021] developed efficiency bounds for aggregating data from multiple logging policies under the reinforcement learning setting. [Carranza and Athey, 2024] extends OPL to multiple source settings by minimizing the worst-case regret over the convex hull of source distributions. However, a key distinction lies in the utilization of target domain information. While their framework seeks a globally robust policy valid for any convex mixture of sources (focusing on external validity), it remains agnostic to the specific covariate geometry of the target domain.

2 Preliminaries

We consider robust offline policy learning from contextual bandit feedback data collected across multiple source sites, together with unlabeled covariates from a target site. We assume there is a finite set of data sources $\mathcal{S}$ from which the dataset is collected, possibly by different known policies in each

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source domain $s \in \mathcal{S}$. Beyond the $\mathcal{S}$ source domains, there is a target domain where we aim to learn a robust policy. Let $G \in \{1, \ldots, S, t\}$ denote the domain (source s or target t).

We consider a multi-source domain adaptation setting involving a set of source domains $\mathcal{S}$ and a target domain t. Let G denote the domain indicator variable. The data generating process within each domain is formulated as a contextual bandit process defined on a common context space $\mathcal{X} \subset \mathbb{R}^p$, a finite action space $\mathcal{A} = \{a_1, \ldots, a_d\}$, and a reward space $\mathcal{Y} \subset \mathbb{R}$.

We adopt the potential outcome framework. For any context X, there exist potential reward outcomes $\{Y(a)\}_{a \in \mathcal{A}}$. The observed reward is $Y = Y(A)$, where A is the action selected by a logging policy. We assume the data within each domain k follows a distribution $\mathcal{D}_k$. For each source $s \in \mathcal{S}$, a labeled dataset $\mathcal{D}_s = \{(X_i^s, A_i^s, Y_i^s)\}_{i=1}^{n_s}$ is collected via a domain-specific behavior policy $\pi_s(a|x)$. For the target domain t, we only observe unlabeled covariates $\{(X_i^t)\}_{i=1}^{n_t}$. Let $n = \sum_{s \in \mathcal{S}} n_s + n_t$ be the total sample size, where we assume n_s and n_t are proportional to n.

Notations. We define the conditional mean outcome function μ_k and the propensity score e_s for domain k and source s respectively, as:

$$\mu_k(a, x) = \mathbb{E}[Y(a) \mid X = x, G = k], \quad e_s(a \mid x) = \mathbb{P}(A = a \mid X = x, G = s).$$

To model the distribution shift, let $q_k = \mathbb{P}(G = k)$ be the marginal domain probability and $s^k(x) = \mathbb{P}(G = k \mid X = x)$ be the conditional domain probability. We define the ratio (importance weight) between the target and source s as: $r_s(x) = s^t(x)/s^s(x)$. The expected value of policy π in the target domain is defined as:

$$V^t(\pi) = \mathbb{E}_{X|G=t}\left[\mu_t(\pi(X), X)\right].$$

Assumption 2.1. For data in each source domain s.

1. (Unconfoundedness): $(Y^s(a_1), \ldots, Y^s(a_d)) \perp\!\!\!\perp A^s \mid X^s$.
2. (Overlap): There exists some $\eta_s > 0$ such that $e_s(a|X) := \mathbb{P}(A^s = a \mid X^s = x) \geq \eta_s$ for any $a \in \mathcal{A}$ and $x \in \mathcal{X}$.
3. (Boundedness): The potential outcomes Y and outcome function μ has a bounded support, i.e., there exists some $B_s > 0$ such that $|Y_s(a)| \leq B_s$ and for all $a \in \mathcal{A}$. Also, there exists a constant C_r such that $|r_s(X)| \leq C_r$.

These assumptions align with the standard offline policy learning frameworks. Specifically, the boundedness condition simplifies our concentration analysis, though our results can be extended to sub-Gaussian outcomes. Regarding the overlap assumption, we acknowledge that recent advances have relaxed this requirement via the pessimism principle [Jin et al., 2025]. However, we enforce uniform overlap here to isolate the challenge of reward heterogeneity. In our multi-source min-max setting, this ensures that the worst-case aggregation addresses genuine conflicting treatment signals across domains rather than compensating for estimation instability caused by disjoint supports.

Assumption 2.2. (Source-mixture outcome transportability) The target domain's reward function is assumed to be a combination of the source domains' reward functions, i.e., $\mu_t(a, X) = \sum_{s=1}^{S} w_s \mu_s(a, X), \quad \sum_{s=1}^{S} w_s = 1, w_s \geq 0$. Additionally, for data in each domain, $0 < \mathbb{P}(G = k|X) < 1$.

Assumption 2.2 constitutes a generalization of the standard transportability assumption widely adopted in the causal inference literature [Stuart et al., 2011, Buchanan et al., 2018, Li et al., 2022, Liu et al., 2025]. The standard condition enforces strict outcome invariance, $(Y(0), \ldots, Y(d)) \perp G|X$, implying that the outcome function $\mu_k(a, X)$ is identical across all domains. This assumption is too strong for heterogeneous multi-source data. In contrast, we relax the transportability assumption by modeling the target mechanism as a convex combination of source mechanisms. This structural relaxation addresses unobservable reward-mechanism drift by defining an identifiable uncertainty set through the observed source mechanisms, while avoiding unrestricted worst-case formulations that would be overly conservative. Similar convex modeling is supported by existing literature [Zhang et al., 2024] and [Wang et al., 2025a], which have been applied to public policy trials and the interpolation of environmental monitoring stations, respectively. The second part of Assumption 2.2 guarantees probabilistic overlap, ensuring that the support of the source covariates covers the target domain.

Assumption 2.3. (Asymptotic estimation rate). For any source site $s \in [S]$, assume that the estimators $\hat{\mu}_s, \hat{e}_s, \hat{r}_s$ for each fold m obtained via cross-fitting satisfy the following convergence rates

$$\|\hat{r}^m - r\|_{\mathbb{P}_s} = o_p(n_s^{-\gamma_r}), \quad \|\hat{e}^m - e\|_{\mathbb{P}_s} = o_p(n_s^{-\gamma_e}), \quad \|\hat{\mu}^m - \mu\|_{\mathbb{P}_s} = o_p(n_s^{-\gamma_\mu})$$

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where the rates satisfy $\gamma_\mu + \gamma_r \geq 1/2$ and $\gamma_\mu + \gamma_e \geq 1/2$ for $\gamma_\mu, \gamma_r, \gamma_e > 0$. $\|f\|_{\mathbb{P}_s} := (\mathbb{E}_{X\sim P_s}[f(X)^2])^{1/2}$.

This assumption is standard in the double machine learning literature [Chernozhukov et al., 2018, Athey and Wager, 2021, Zhou et al., 2023]. It can be satisfied by employing a range of flexible machine learning methods. For example, it can be achieved by estimators at a rate faster than $n^{-1/4}$ under mild conditions.

3 Formulation

Lemma 3.1. *Suppose Assumption 2.1 and Assumption 2.2 hold, we have* $V^t(\pi) = \sum_{s=1}^S w_s V^{s\to t}(\pi)$ *where* $V^{s\to t}(\pi)$ *is defined as follows:*

$$V^{s\to t}(\pi) := \mathbb{E}[\frac{\mathbb{I}(G=t)}{q_t}\mu_s(\pi(X),X)] + \mathbb{E}\left[\frac{\mathbb{I}(G=s)}{q_t}r_s(X)\frac{\mathbb{I}(A=\pi(X))}{e_s(A|X)}(Y-\mu_s(A,X))\right]$$

The proof is deferred to Appendix A.1. This lemma establishes a constructive identification strategy that effectively integrates information from both domains. Specifically, the first term utilizes unlabeled target covariates for model-based imputation, while the second term leverages labeled source data for bias correction via importance weighting. This decomposition provides a tractable form for constructing the doubly robust estimator required in our subsequent optimization.

Based on Assumption 2.2, the value of a policy π in the target domain, denoted as $V^t(\pi) := \mathbb{E}_{X\sim P_t}[\mu_t(X,\pi(X))]$, can be decomposed into a convex combination of source-specific values adapted to the target population:

$$V^t(\pi) = \mathbb{E}_{X\sim P_t}\left[\sum_{s=1}^S w_s\mu_s(X,\pi(X))\right] = \sum_{s=1}^S w_s V^{s\to t}(\pi) \tag{1}$$

where $V^{s\to t}(\pi) := \mathbb{E}_{X\sim P_t}[\mu_s(X,\pi(X))]$. It represents the expected reward from the reward mechanism of source s.

Since the target mixture weights are unknown and not identifiable from unlabeled target covariates alone, we adopt a minimax source-mixture robust criterion. We define the robust value function $Q(\pi)$ as the worst-case performance over all possible valid weight assignments in the simplex Δ_{S-1}:

$$Q(\pi) := \min_{w\in\Delta_{S-1}} \sum_{s=1}^S w_s V^{s\to t}(\pi). \tag{2}$$

where $w \in \Delta_{S-1} = \{w \in \mathbb{R}^S \mid \sum_{s=1}^S w_s = 1, \min_s w_s \geq 0\}$.

Our objective is to learn an optimal policy π^* that maximizes this robust value:

$$\pi^* = \arg\max_{\pi\in\Pi} Q(\pi) = \arg\max_{\pi\in\Pi} \min_{w\in\Delta_{S-1}} \left\{\sum_{s=1}^S w_s V^{s\to t}(\pi)\right\}. \tag{3}$$

Observing that the inner minimization problem is linear with respect to w, the minimum is attained at one of the vertices of the simplex. Thus, the optimization over Δ_{S-1} simplifies to a discrete minimization over the sources:

$$\pi^* = \arg\max_{\pi\in\Pi} \min_{s\in[S]} V^{s\to t}(\pi). \tag{4}$$

Directly optimizing the objective in Equation (4) is computationally intractable for gradient-based methods due to the non-differentiability of the $\min$ operator. To address this, we construct a differentiable surrogate objective by approximating the hard minimum with the *Softmin* operator (Log-Sum-Exp) using a temperature parameter $\tau > 0$. The optimization problem is reformulated as maximizing the following smoothed objective $\mathcal{L}_\tau(\pi)$:

$$\max_{\pi\in\Pi} \mathcal{L}_\tau(\pi), \quad \mathcal{L}_\tau(\pi) = -\tau\log\left(\frac{1}{S}\sum_{s=1}^S \exp\left(-V^{s\to t}(\pi)/\tau\right)\right). \tag{5}$$

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It is well-established that $\lim_{\tau\to 0}\mathcal{L}_\tau(\pi) = \min_s \hat{V}^{s\to t}(\pi)$, ensuring that the solution to this smooth problem converges to the robust policy.

To empirically solve Equation (5), we require an estimator for each component $V^{s\to t}(\pi)$. Following lemma A.1, we propose a robust estimator that leverages the unlabeled target data and labeled source data. For any source s, the estimator is given by:

$$\hat{V}^{s\to t}(\pi) := \frac{1}{n_t}\sum_{i\in\mathcal{D}_t}\hat{\mu}_s(\pi(X_i),X_i) + \frac{1}{n_t}\sum_{i\in\mathcal{D}_s}\hat{r}_s(X_i)\frac{\mathbb{I}(A_i=\pi(X_i))}{\hat{e}_s(A_i,X_i)}\big(Y_i-\hat{\mu}_s(A_i,X_i)\big) \tag{6}$$

where $\hat{r}_s, \hat{\mu}_s, \hat{e}_s$ are nuisance functions estimated from source s.

Also, we define the multiple source robust estimator for a given policy π as

$$\hat{V}^t(\pi) = \sum_{s=1}^{S} w_s \hat{V}^{s\to t}(\pi) \tag{7}$$

The following theorem states that the estimated policy value $\hat{V}^t(\pi)$ is consistent for estimating $V^t(\pi)$ and is asymptotically normal.

Theorem 3.2. *(Asymptotic Normality) Under Assumption 2.1, Assumption 2.2, and Assumption 2.3, the multi-source doubly robust estimator $\hat{V}^t(\pi)$ defined in Equation* (7) *satisfies*

$$\sqrt{n}\big(\hat{V}^t(\pi) - V^t(\pi)\big) \xrightarrow{d} \mathcal{N}\big(0,\sigma^2_{\text{total}}\big), \quad \sigma^2_{\text{total}} = \sum_{s=1}^{S}\frac{w_s^2}{q_s}\sigma^2_{s\to t}$$

and $\sigma^2_{s\to t}$ is

$$\sigma^2_{s\to t} = \frac{q_s}{q_t}\text{Var}_t\big(\mu_s(\pi(X),X)\big) + \frac{q_s^2}{q_t^2}\text{Var}_s(\psi_s(Z))$$

$\text{Var}_t(\text{Var}_s)$ *means the variance is taken at the target domain t (source domain s). Function ψ is defined as:* $\psi_s(Z) = \frac{r_s(X)\,\mathbb{I}(A=\pi(X))}{e_s(A|X)}\big(Y-\mu_s(A,X)\big)$.

The proof of Theorem 3.2 is deferred to the appendix. This optimality arises because our site-specific doubly robust scores structurally align with the unique efficient influence function for covariate shift characterized by Liu et al. [2025], while our aggregation strategy parallels the efficient inference framework for multiple logging policies established by Kallus et al. [2021]. Our method inherits these favorable asymptotic properties, ensuring statistical optimality even in the presence of nuisance estimation errors.

3.1 Regret Analysis

Let π^* denote the optimal robust policy that maximizes the worst-case value defined in Equation (3). We define the multi-source regret $R(\pi)$ as:

$$R(\pi) := Q(\pi^*) - Q(\pi). \tag{8}$$

To establish uniform convergence bounds, we characterize the complexity of the policy class Π using the *Hamming entropy integral*. This measure captures the complexity of the policy space based on the discrepancies in decision-making.

Definition 3.3 (Hamming Entropy Integral)**.** Given a policy class Π and n data points $\{x_1,...,x_n\} \subseteq \mathcal{X}$, The ε-covering number of $\{x_1,\ldots,x_n\}$, denoted as $\mathcal{C}(\varepsilon,\Pi;\{x_1,\ldots,x_n\})$, is the smallest number L of policies $\{\pi_1,\ldots,\pi_L\}$ in Π, such that $\forall\pi\in\Pi$, $\exists\pi'_\ell$ such that $d_H(\pi,\pi_\ell)\le\varepsilon$.

The Hamming distance between two policies $\pi_1,\pi_2\in\Pi$ is $d_H(\pi_1,\pi_2) := \frac{1}{n}\sum_{i=1}^n \mathbb{I}\{\pi_1(X_i)\neq \pi_2(X_i)\}$. The Hamming entropy integral of Π is defined as $\kappa(\Pi) := \int_0^1\sqrt{\log N_H(\varepsilon^2,\Pi)}\,d\varepsilon$, where $N_H(\varepsilon,\Pi) := \sup_{n\ge1}\sup_{x_1,\ldots,x_n}\mathcal{C}(\varepsilon,\Pi;\{x_1,\ldots,x_n\})$.

Let $\hat{\pi}$ be the empirical policy obtained by solving the maximization problem in Equation (5) as $\tau\to 0$ (i.e., $\arg\max_\pi \min_s \hat{V}^{s\to t}(\pi)$). The following theorem provides a regret upper bound for the learned policy.

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Theorem 3.4 (Regret Bound)**.** *Let $\hat{\pi}$ be the policy obtained by solving Equation* (5)*, or equivalently the limit of the soft-min objective as $\tau \to 0$. Under Assumption 2.1, Assumption 2.2, and Assumption 2.3, with probability at least $1-\beta$:*

$$R(\hat{\pi}) \leq 2 \max_{s \in [S]} \left[B_s (1 + \frac{2C_r}{\eta_s} \sqrt{\frac{n_s}{n_t}}) \right] \cdot \frac{1}{\sqrt{n_t}} \left(96\kappa(\Pi) + \sqrt{2\log(4S/\beta)} \right)$$

The constants B_s, C_r, η_s are denoted in Assumption 2.1.

The proof of Theorem 3.4 is deferred to the appendix. The regret decays at the rate of $O(1/\sqrt{n_t})$. However, the multi-site setting introduces a price of robustness. consider a balanced asymptotic regime where the total sample size n is evenly distributed across S source sites (i.e., $n_s \asymp n_t \asymp n/S$). In this case, the bound simplifies to: $R(\hat{\pi}) = O_p \left(\sqrt{\frac{S}{n}} \left(\kappa(\Pi) + \sqrt{\log S} \right) \right)$. It shows that fragmentation of data into more sites increases the difficulty of the worst-case optimization. Nevertheless, the logarithmic dependence on S indicates that our method maintains efficiency even as the number of source domains grows.

4 Algorithm

We now present the practical optimization framework for Robust Multi-source OPL, which is outlined in Algorithm 1.

To construct the Multi-Source Doubly Robust estimator defined in Equation (6), we require estimates of the nuisance components: the outcome models $\hat{\mu}_s$, propensity scores $\hat{e}_s$, and density ratios $\hat{r}_s$. In step 1, we employ the cross-fitting method to mitigate overfitting bias. This procedure effectively decouples nuisance learning from policy evaluation, ensuring asymptotic normality [Chernozhukov et al., 2018]. The nuisance terms are pre-calculated and fixed prior to the optimization loop.

With the nuisances fixed, we aim to maximize the worst-case value. To enable gradient-based updates, we relax the discrete policy indicator function $\mathbb{I}(A = \pi(X))$ using a differentiable Softmax approximation: $\pi_\theta(a|x) = \exp(f_\theta(x,a)) / \sum_{a' \in \mathcal{A}} \exp(f_\theta(x,a'))$. The function $f_\theta(x,a)$ can be instantiated flexibly as a linear model ($f_\theta(x,a) = \theta_a^\top x$) or as differentiable nonlinear models. We treat this smooth relaxation during training, and ultimately deploy the induced deterministic policy by taking the $\arg\max$ over actions.

We adopt a smooth surrogate objective $\mathcal{L}_\tau$ using the *Soft-min* operator in Equation (5):

$\max_\theta -\tau \log \left(\frac{1}{S} \sum_{s=1}^S \exp \left(-\hat{V}^{s \to t}(\pi_\theta)/\tau \right) \right)$. The parameter τ serves to tune the sensitivity of the possible bad site: $\tau \to 0$ approximates the hard max-min (worst-case protection), while $\tau \to \infty$ recovers the arithmetic mean $\frac{1}{S} \sum_{s=1}^S V^{s \to t}(\pi)$. We update θ by gradient ascent: $\theta^{(j+1)} \leftarrow \theta^{(j)} + \eta \nabla_{\mathcal{L}_\tau}(\pi)$ where $\nabla_{\mathcal{L}_\tau}(\pi) = \sum_{s=1}^S \lambda_s(\theta) \nabla_\theta \hat{V}^{s \to t}(\theta)$ and $\lambda_s(\theta) = \exp \left(-\hat{V}^{s \to t}(\theta)/\tau \right) / \sum_{k=1}^S \exp \left(-\hat{V}^{k \to t}(\theta)/\tau \right)$.

Definition 4.1. (KL divergence) The KL divergence between two distributions Q and P is defined as

$$D_{\mathrm{KL}}(Q \,\|\, P) = \mathbb{E}_Q \left[\log \frac{dQ}{dP} \right],$$

where $\frac{dQ}{dP}$ is the Radon-Nikodym derivative of Q with respect to P.

Proposition 4.2. *(Equivalence of Soft-min and KL-Regularized Objective) The Soft-min objective function $\mathcal{L}_\tau(\pi)$ defined in Equation* (5) *is mathematically equivalent to the value of a robust objective with a KL-divergence penalty on the target mixture weights w:*

$$\mathcal{J}_\tau(\pi) = \min_{w \in \Delta_{S-1}} \{ \sum_{s=1}^S w_s V^{s \to t}(\pi) + \tau \cdot D_{\mathrm{KL}}(w \| u) \} \tag{9}$$

where Δ_{S-1} is the probability simplex, $u = [1/S, \ldots, 1/S]$ is the uniform distribution over source sites, and $D_{\mathrm{KL}}(w\|u) = \sum_s w_s \log(S w_s)$ is the KL divergence between w and u. In other words, the optimization problem in Equation (5) *is equivalent to solving: $\max_\pi \mathcal{J}_\tau(\pi)$.*

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Algorithm 1 Multi-Source Robust Policy Learning

Input: Source datasets $\{\mathcal{D}_s\}_{s=1}^{S}$, target covariates $\mathcal{D}_t$, folds M, temperature τ, learning rate η.
1. Cross-Fitted Nuisance Estimation:
for each source $s = 1, \ldots, S$ and target t **do**
  Randomly split $\mathcal{D}_s \cup \mathcal{D}_t$ into M folds $\{\mathcal{I}_{s,m}\}_{m=1}^{M}$.
  for each fold $m = 1, \ldots, M$ **do**
    Estimate $\hat{\mu}_{s,(-m)}$, $\hat{e}_{s,(-m)}$, and $\hat{r}^{s\to t}_{(-m)}$ using data excluding fold m.
  end for
end for
2. Initialization: Initialize policy parameters $\theta^{(0)}$.
3. Iterative Optimization:
for $j = 0, 1, \ldots$ until convergence **do**
  a. Compute $\widehat{V}_{s\to t}(\theta^{(j)})$ using the pre-computed cross-fitted estimates and the policy function.
  b. Compute soft-min weights:
  $\lambda_s \propto \exp(-\widehat{V}^{s\to t}(\theta^{(j)})/\tau)$.
  c. Update θ via gradient ascent on the weighted objective:
    $\theta^{(j+1)} \leftarrow \theta^{(j)} + \eta \sum_{s=1}^{S} \lambda_s \nabla_\theta \widehat{V}_{s\to t}(\theta^{(j)})$.
end for
4. Output: The learned policy $\pi_\theta(x)$.

The proof of proposition 4.2 is deferred to the appendix. Although the algorithm optimizes a smoothed soft-min objective for computational convenience, this formulation shows that this smoothing has an adversarial source-weighting interpretation. The adversary chooses mixture weights over source-specific target-adapted values but is penalized for deviating from the uniform mixture.

Here, τ serves as a knob regulating the learner's conservatism. As $\tau \to 0$, the mechanism approaches "absolute pessimism," where the weight distribution collapses to the single worst-performing source, recovering the hard max-min guarantee. As $\tau \to \infty$, the mechanism approaches "risk neutrality" where weights converge to the uniform distribution. By tuning τ, our framework bridges these extremes, trading off the strict safety of the max-min solution against the estimation power obtained from aggregating multiple sources.

5 Experiments

5.1 Experimental Setup

Data-generating process. We consider a binary-action contextual bandit with $S = 3$ source environments and an unlabeled target covariate sample. Each context has dimension $d = 8$. Source and target covariates follow Gaussian distributions with identity covariance and mild mean shifts:

$$X \mid G = s \sim \mathcal{N}(m_s, I_d), \qquad X_t \sim \mathcal{N}(m_t, I_d),$$

where $m_1 = (-0.4, 0, 0, 0, \ldots, 0)$, $m_2 = (0, -0.4, 0, 0, \ldots, 0)$, $m_3 = (0, 0, -0.4, 0, \ldots, 0)$, and $m_t = (0.2, 0.2, 0.2, 0, \ldots, 0)$. Covariates are observed by the learner. The source potential outcomes follow the reward model. For source s, $\mu_s(0, x) = b_s + u^\top x$, and $\mu_s(1, x) = \mu_s(0, x) + \eta_s\, \sigma(10(x_{j_s} + 0.5))$, where $\sigma(z) = 1/(1 + \exp(-z))$, $j_s = s$ for $s \in \{1, \cdots, S\}$, and multipliers are $(\eta_1, \eta_2, \eta_3) = (-8, 3, 5)$. The slope parameter is generated once as $\tilde{u} \sim \mathcal{N}(0, I_8), u = 0.5\, \frac{\tilde{u}}{\|\tilde{u}\|_2}$. The source-specific intercepts are $b_s = 0.1\, U_s, U_s \sim \mathrm{Unif}(0, 1)$.

For each source, logged actions are generated by a weakly source-specific behavior policy. The logging probability is $\tilde{e}_s(1 \mid x) = \frac{\exp(h_s(x)/T)}{1+\exp(h_s(x)/T)}$ with behavior temperature $T = 2.0$., and the raw behavior score is linear: $h_s(x) = 0.5\, x_s + \alpha_s$ with $\alpha_s \in \{-0.2, 0, 0.2\}$. We enforce overlap with a minimum probability of $p_{\min} = 0.1$ as $e_s(1 \mid x) = (1 - 2p_{\min})\tilde{e}_s(1 \mid x) + p_{\min}, e_s(0 \mid x) = 1 - e_s(1 \mid x)$. The logged action is sampled as $A \sim \mathrm{Bernoulli}(e_s(1 \mid X))$, and the observed reward is $Y = \mu_s(A, X) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 0.2^2)$.

Training. For each of the 100 random seeds, we generate a logged source sample with 3000 samples per source. The target sample is split into 2000 unlabeled covariates used during nuisance

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estimation/training and 1000 held-out covariates used only for evaluation. We adopt linear stochastic policies and logistic regression nuisance models with three-fold cross-fitting. For our proposed approach, we set the learning rate to 5×10^{-3}. The proposed multi-source robust policy (MS) optimizes a smooth approximation of the worst-source objective using soft-min temperatures $\tau \in \{0.5, 1.0, 5.0\}$ as presented in Algorithm 1 . A smaller τ here is expected to be more conservative; a larger τ places more weight on average performance.

Simplex grid evaluation. The target reward mechanism is simulated to lie in the convex hull of the source reward mechanisms: $\mu_w(a, x) = \sum_{s=1}^{S} w_s \mu_s(a, x), w \in \Delta$ as shown in Assumption 2.2. After training, we evaluate policies from different methods on a uniform grid over the simplex with a step size 0.1, yielding 66 target mixtures. For each held-out target context and each grid point w, the target value is computed using the true source reward functions. For each grid point, we first average policy values across random seeds and then summarize the resulting value surface over grid points, including the minimum policy value, bottom-10%, mean value, standard deviation, and robust regret. Here, bottom-10% denotes the average value over the lowest 10% of simplex-grid target mixtures. The reported robust regret is the empirical version of our definition in Equation 8.

We compare our method against two non-robust approaches. The first adopts a single-source covariate shift doubly robust policy-learning objective, sharing the same core design as a recent optimal policy adaptation method in [Liu et al., 2025]. We name it CS (short for covariate shift) in the following tables. The second is a standard off-policy learning baseline implemented with Open Bandit Pipeline (OBP) [Saito et al., 2021], which does not explicitly use unlabeled target covariates for covariate-shift correction. Since both approaches are single-source policy-learning approaches, we instantiate each of them in two ways: 1. **Pooled**: merge all source data and train a single policy on the pooled sample. 2. **AvgSingle**: train one policy per source and report the average value of the source-specific policies.

5.2 Performance Comparison

Table 1 reports the main comparison among our multi-source method (MS) and the baselines. Our method substantially improves the downside region and obtains the highest minimum policy value, highest bottom 10% policy values and lowest regret across the simplex. As τ goes from 0.5 to 5.0, our method achieves a higher mean value, and its lower tail is less conservative. This illustrates the expected robust-average performance tradeoff controlled by the parameter τ in our theory. The table shows that no pooled source policy or average source-specific policy is reliable over the simplex. This supports the motivation that a robust multi-source policy is required when the target mixture is unknown.

Table 1: Simplex-grid policy value and robust regret.

Method	Policy Value				Robust Regret
	Min	Bottom 10%	Mean	Std.	
MS(τ=0.5)	**0.080**	**0.158**	0.433	0.152	0.009
MS(τ=1.0)	0.047	0.147	0.497	0.194	0.042
MS(τ=5.0)	-0.191	-0.022	0.563	0.319	0.280
CS-Pooled	-2.842	-2.091	0.462	1.366	2.931
CS-AvgSingle	-3.938	-3.005	0.112	1.649	4.027
OBP-Pooled	-2.193	-1.575	0.515	1.114	2.283
OBP-AvgSingle	-3.864	-2.942	0.135	1.628	3.953

Figure 1 visualizes the policy value surface over the simplex. Each triangular panel shows the policy value over the simplex, where each point corresponds to a target reward mixture w. The MS (our method with $\tau = 0.5$) surface is much flatter and avoids the low-value regions visible for the baselines. The baselines can perform well in favorable regions of the simplex, but they collapse near mixtures where the harmful source receives substantial weight. In contrast, our method sacrifices some favorable-region value in exchange for substantially better lower-tail performance.

5.3 Ablation Study

Source number scaling. We evaluate whether the proposed method remains stable as the number of source increases. We consider $S \in \{4, 6, 8\}$ with a fixed total source sample size $n = 9000$, so each source receives fewer samples as S grows. As the dimension of the full simplex increases, we

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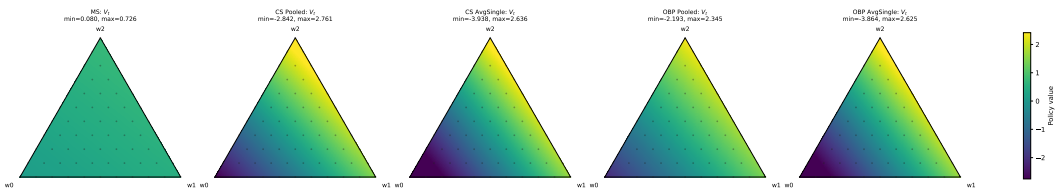


Figure 1: Policy-value surfaces over the source-weight simplex.

evaluate policies on pairwise edge grids, which probe target mixtures between every pair of sources. Results in Table 2 suggest that the robust objective continues to maintain the highest lower-tail values as the number of sources increases.

Table 2: Policy value under scaling source numbers.

Number of sources	Method	Policy Value				Robust Regret
		Min	Bottom 10%	Mean	Std.	
4	MS(τ=0.5)	**0.064**	**0.070**	0.154	0.058	0.034
	CS-Pooled	-0.417	-0.337	0.246	0.355	0.515
	CS-AvgSingle	-2.919	-2.614	-0.276	1.449	3.017
	OBP-Pooled	-0.262	-0.243	0.190	0.312	0.367
	OBP-AvgSingle	-2.900	-2.596	-0.262	1.448	2.995
6	MS(τ=0.5)	**0.069**	**0.070**	0.117	0.048	0.025
	CS-Pooled	-0.526	-0.423	0.173	0.387	0.690
	CS-AvgSingle	-2.952	-2.501	-0.167	1.359	3.037
	OBP-Pooled	-0.470	-0.409	0.150	0.381	0.570
	OBP-AvgSingle	-2.914	-2.469	-0.156	1.348	2.998
8	MS(τ=0.5)	**0.046**	**0.136**	0.266	0.090	0.063
	CS-Pooled	-4.723	-2.675	0.336	1.549	3.348
	CS-AvgSingle	-3.717	-2.787	0.209	1.546	3.738
	OBP-Pooled	-2.145	-1.691	0.377	1.137	2.337
	OBP-AvgSingle	-3.498	-2.748	0.207	1.525	3.690

Source data imbalance. We conduct experiments to validate the performance of our method under imbalanced multi-source data settings. The total sample size is fixed at $n = 9000$. The result in Table 3 shows that our method remains stable under substantial source sample imbalance: its minimum and bottom-10% policy values stay close across all imbalanced settings, indicating that the learned robust policy does not collapse when one source has fewer samples. Although the mean value is lower than in the balanced case, the tail performance remains controlled. More ablation study details are provided in the appendix.

Table 3: Policy value under imbalanced source sample sizes.

Source data sizes	Policy Value			
	Min	Bottom 10%	Mean	Std.
$3000, 3000, 3000$	0.080	0.158	0.433	0.152
$1500, 3750, 3750$	0.086	0.123	0.252	0.070
$4125, 4125, 750$	0.086	0.120	0.239	0.065
$750, 4125, 4125$	0.085	0.126	0.267	0.078

6 Conclusion

In this paper, we present a robust offline policy learning method that generalizes from multiple source sites to an unlabeled target site. By optimizing the worst-case value over the convex hull of source domains, our method establishes a performance guarantee against distributional shifts. We establish the asymptotic normality and minimax regret bounds of our estimator, validating its statistical efficiency. Empirically, our approach exhibits stability compared to the baselines. However, our approach is limited by the convex transportability assumption. Future work will focus on relaxing this requirement and extending the framework to sequential decision-making settings.

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A Lemmas and Propositions

Lemma A.1. *Suppose Assumption 2.1 and Assumption 2.2 hold, we have* $V^t(\pi) = \sum_{s=1}^S w_s V^{s\to t}(\pi)$ *where* $V^{s\to t}(\pi)$ *is defined as follows:*

$$V^{s\to t}(\pi) := \mathbb{E}[\frac{\mathbb{I}(G=t)}{q_t}\mu_s(\pi(X),X)] + \mathbb{E}\left[\frac{\mathbb{I}(G=s)}{q_t}r_s(X)\frac{\mathbb{I}(A=\pi(X))}{e_s(A|X)}(Y-\mu_s(A,X))\right]$$

Lemma A.2. *(Theorem 4.10 in [Wainwright, 2019]). If* $f(z)\in[-B,B]$*, we have a probability of at least* $1-\exp\left(-\frac{n\epsilon^2}{2B^2}\right)$*,*

$$\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n f(z_i)-\mathbb{E}[f(z)]\right|\le 2\mathcal{R}_n(\mathcal{F})+\epsilon.$$

Lemma A.3. *(Dudley's Theorem, (5.48) in [Wainwright, 2019]). If* $f(z)\in[-B,B]$*, we have a bound for the Rademacher complexity,*

$$\mathcal{R}_n(\mathcal{F})\le\frac{24}{\sqrt{n}}\int_0^{2B}\sqrt{\log N(t,\mathcal{F},\|\cdot\|_{L^2(P_n)})}\,dt,$$

where $N(t,\mathcal{F},\|\cdot\|_{L^2(P_n)})$ *is the* t*-covering number of set* $\mathcal{F}$ *and the metric* $\|\cdot\|_{L^2(P_n)}$ *is defined by*

$$\|f-g\|_{L^2(P_n)}=\sqrt{\frac{1}{n}\sum_{i=1}^n(f(z_i)-g(z_i))^2}.$$

Lemma A.4. *[lemma D.2 in [Wang et al., 2025a]] Let* $\{\sigma_i\}_{i=1}^n$ *be i.i.d. Rademacher random variables (i.e.,* $\mathbb{P}(\sigma_i=1)=\mathbb{P}(\sigma_i=-1)=1/2$*), and let* $\mathcal{F}$ *be a family of functions such that each* $f\in\mathcal{F}$ *satisfies* $|f(Z_i)|\le c_i$ *for some constants* $c_i\ge 0$*. Let* $\kappa(\mathcal{F})$ *denote the Dudley entropy integral of* $\mathcal{F}$*. Then,*

$$\mathbb{E}\left[\sup_{f\in\mathcal{F}}\left|\frac{1}{n}\sum_{i=1}^n\sigma_i f(Z_i)\right|\right]\le\frac{\sqrt{\sum_{i=1}^n c_i^2}}{n}\cdot C\left(\kappa(\mathcal{F})\right)$$

where $C\left(\kappa(\mathcal{F})\right)$ *is a constant depending only on the entropy integral* $\kappa(\mathcal{F})$ *(e.g.,* $C(\kappa)=32+4\kappa$ *in typical settings).*

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A.1 Proof of lemma A.1

The target policy value is defined as:

$$V^t(\pi(X)) = \mathbb{E}_{X|G=t}\left[\sum_a \pi(a|X)\mu_t(a,X)\right].$$

For a deterministic policy, this simplifies to $\mathbb{E}_{X|G=t}[\mu_t(\pi(X),X)]$.

By Assumption 2.2:

$$V^t(\pi) = \mathbb{E}_{X|G=t}\left[\sum_{s=1}^{S} w_s\mu_s(\pi(X),X)\right] = \sum_{s=1}^{S} w_s\mathbb{E}_{X|G=t}\left[\mu_s(\pi(X),X)\right]. \tag{10}$$

For each source s, consider the term $\mathbb{E}_{X|G=t}[\mu_s(\pi(X),X)]$. Using the density ratio, we express it in terms of the source distribution:

$$\mathbb{E}_{X|G=t}[\mu_s(\pi(X),X)] = \int \mu_s(\pi(x),x)\, p(x|G=t)\, dx.$$

Note that:

$$p(x|G=t) = \frac{s^t(x)p(x)}{q_t}, \quad p(x|G=s) = \frac{s^s(x)p(x)}{q_s},$$

hence

$$p(x|G=t) = \frac{s^t(x)q_s}{s^s(x)q_t}p(x|G=s) = r_s(x)\frac{q_s}{q_t}p(x|G=s).$$

Therefore,

$$\mathbb{E}_{X|G=t}[\mu_s(\pi(X),X)] = \mathbb{E}_{X|G=s}\left[r_s(X)\frac{q_s}{q_t}\mu_s(\pi(X),X)\right]. \tag{11}$$

Under Assumption 2.1 (ignorability and overlap) for each source s, we have:

$$\mu_s(a,X) = \mathbb{E}\left[\frac{\mathbb{I}(A=a)}{e_s(a|X)}Y \,\middle|\, X, G=s\right].$$

In particular, for $a=\pi(X)$:

$$\mu_s(\pi(X),X) = \mathbb{E}\left[\frac{\mathbb{I}(A=\pi(X))}{e_s(A|X)}Y \,\middle|\, X, G=s\right]. \tag{12}$$

Substituting Equation (12) into Equation (11), we have:

$$\mathbb{E}_{X|G=t}[\mu_s(\pi(X),X)] = \mathbb{E}_{X|G=s}\left[r_s(X)\frac{q_s}{q_t}\mathbb{E}\left[\frac{\mathbb{I}(A=\pi(X))}{e_s(A|X)}Y \,\middle|\, X, G=s\right]\right].$$

By the law of iterated expectations and $q_s := \mathbb{P}(G=s)$

$$\mathbb{E}_{X|G=t}[\mu_s(\pi(X),X)] = \mathbb{E}\left[\frac{\mathbb{P}(G=s)}{q_t}r_s(X)\frac{\mathbb{I}(A=\pi(X))}{e_s(A|X)}Y \middle| G=s\right].$$

By $\mathbb{E}[Z|G=s] = \mathbb{E}[\mathbb{I}(G=s)Z]/\mathbb{P}(G=s)$, we then get:

$$\mathbb{E}_{X|G=t}[\mu_s(\pi(X),X)] = \mathbb{E}\left[\frac{\mathbb{I}(G=s)}{q_t}r_s(X)\frac{\mathbb{I}(A=\pi(X))}{e_s(A|X)}Y\right]. \tag{13}$$

Note that the equality $\mathbb{E}[Z|G=s] = \mathbb{E}[\mathbb{I}(G=s)Z]/\mathbb{P}(G=s)$ is from

$$\begin{aligned}
\mathbb{E}[\mathbb{I}(G=s)Z] &= \sum_g\sum_z z\cdot\mathbb{I}(g=s)\mathbb{P}(G=g,Z=z)\\
&= \sum_z z\cdot\mathbb{P}(G=s,Z=z)\\
&= \sum_z z\cdot\mathbb{P}(G=s)\cdot\mathbb{P}(Z=z|G=s)\\
&= \mathbb{P}(G=s)\mathbb{E}[Z|G=s]
\end{aligned}$$

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Equation (13) is the inverse probability weighting (IPW) identification. To obtain the doubly robust form, we add a term with a zero mean:

$$\mathbb{E}\left[\frac{\mathbb{I}(G=s)}{q_t} r_s(X) \frac{\mathbb{I}(A=\pi(X))}{e_s(A|X)}(Y-\mu_s(A,X))\right] = 0, \tag{14}$$

because $\mathbb{E}[Y-\mu_s(A,X)|X,A,G=s]=0$. Hence, we add this zero mean term to Equation (10) without changing it.

Combining Equation (10) and Equation (14), we define:

$$V^{s\to t}(\pi) = \mathbb{E}[\frac{\mathbb{I}(G=t)}{q_t}\mu_s(\pi(X),X)] + \mathbb{E}\left[\frac{\mathbb{I}(G=s)}{q_t} r_s(X) \frac{\mathbb{I}(A=\pi(X))}{e_s(A|X)}(Y-\mu_s(A,X))\right].$$

Therefore, we have $V^t(\pi) = \sum_{s=1}^S w_s V^{s\to t}(\pi)$.

Define the estimator:

$$\hat{V}^{s\to t}(\pi) := \frac{1}{n_t}\sum_{i\in\mathcal{D}_t}\hat{\mu}_s(\pi(X_i),X_i) + \frac{1}{n_t}\sum_{i\in\mathcal{D}_s}\hat{r}(X_i)\frac{\mathbb{I}(A_i=\pi(X_i))}{\hat{e}_s(A_i,X_i)}\big(Y_i-\hat{\mu}_s(A_i,X_i)\big)$$

This derivation provides the identification of the doubly robust estimator.

A.2 Proof of Proposition 4.2

Proof. We prove the equivalence using the method of Lagrange multipliers. By Definition 4.1 (the definition of KL-divergence), we have $D_{\mathrm{KL}}(w\|u) = \sum_s w_s \log(Sw_s)$. Let $J(w) = \sum_{s=1}^S w_s V^{s\to t} + \tau\sum_{s=1}^S w_s\log(Sw_s)$ be the primal objective. We introduce a Lagrange multiplier ν for the simplex constraint $\sum_{s=1}^S w_s = 1$. The Lagrangian is given by:

$$L(w,\nu) = \sum_{s=1}^S w_s V^{s\to t} + \tau\sum_{s=1}^S w_s(\log w_s + \log S) + \nu\left(1-\sum_{s=1}^S w_s\right). \tag{15}$$

Taking the derivative with respect to w_s and setting it to zero yields the KKT condition:

$$\frac{\partial L}{\partial w_s} = V^{s\to t} + \tau(\log w_s + 1 + \log S) - \nu = 0. \tag{16}$$

Rearranging the KKT condition to isolate w_s:

$$\log w_s = \frac{\nu - V^{s\to t}}{\tau} - (1+\log S) \implies w_s^* = \exp\left(\frac{\nu}{\tau} - 1 - \log S\right)\exp\left(-\frac{V^{s\to t}}{\tau}\right). \tag{17}$$

Let $Z(\nu) = \exp(\frac{\nu}{\tau} - 1 - \log S)$ be the term independent of s. We write $w_s^* = Z(\nu)\exp(-V^{s\to t}/\tau)$. Apply the constraint $\sum_{s=1}^S w_s^* = 1$:

$$Z(\nu)\sum_{s=1}^S \exp\left(-\frac{V^{s\to t}}{\tau}\right) = 1 \implies Z(\nu) = \frac{1}{\sum_{k=1}^S \exp(-V^{k\to t}/\tau)}. \tag{18}$$

Substituting $Z(\nu)$ back into Equation (17), we recover the equation matching λ_s in our algorithm:

$$w_s^* = \frac{\exp(-V^{s\to t}/\tau)}{\sum_{k=1}^S \exp(-V^{k\to t}/\tau)}. \tag{19}$$

We substitute w^* back into the primal objective $J(w)$. From the KKT condition in Equation (16), we know that $V^{s\to t} + \tau\log(Sw_s^*) = \nu - \tau$. Thus:

$$J(w^*) = \sum_{s=1}^S w_s^*\left(V^{s\to t} + \tau\log(Sw_s^*)\right) = \sum_{s=1}^S w_s^*(\nu-\tau) = \nu - \tau. \tag{20}$$

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To find the value of $\nu - \tau$, we take the logarithm of the expression for $Z(\nu)$:

$$\log Z(\nu) = \frac{\nu}{\tau} - 1 - \log S \implies \nu - \tau = \tau(\log Z(\nu) + \log S). \tag{21}$$

Substitute Equation (21) into Equation (20) and use Equation (18):

$$\begin{aligned} J(w^*) &= \nu - \tau \\ &= \tau(\log Z(\nu) + \log S) \\ &= \tau \left(-\log \left(\sum_{k=1}^{S} \exp(-V^{k\to t}/\tau) \right) + \log S \right) \\ &= -\tau \log \left(\frac{1}{S} \sum_{k=1}^{S} \exp\left(-\frac{V^{k\to t}}{\tau} \right) \right). \end{aligned}$$

This confirms that the optimal value of the KL-regularized problem is exactly the soft-min objective $\mathcal{L}_\tau(\pi)$ in Equation (5).

□

B Proof of Theorem 3.2

Proof. **Step 1: Cross-Fitting Setup**

Partition the data into M folds. For each fold $m = 1, \ldots, M$, we have data $\mathcal{D}_s^m$ and $\mathcal{D}_t^m$. The training set is $\mathcal{D}^{-m} = (\mathcal{D}_s \setminus \mathcal{D}_s^m) \cup \mathcal{D}_t^m$. We train the nuisance estimators $\hat{\eta}^m = (\hat{r}_s^m, \hat{e}_s^m, \hat{\mu}_s^m)$ on $\mathcal{D}^{-m}$.

The fold-m source-to-target estimator is:

$$\hat{V}_m^{s\to t}(\pi) = \frac{1}{n_{t,m}} \sum_{i \in \mathcal{D}_t^m} \hat{\mu}_s^m(\pi(X_i), X_i) + \frac{n_s}{n_t} \frac{1}{n_{s,m}} \sum_{i \in \mathcal{D}_s^m} \hat{r}_s^m(X_i) \frac{\mathbb{I}(A_i = \pi(X_i))}{\hat{e}_s^m(A_i, X_i)} (Y_i - \hat{\mu}_s^m(A_i, X_i)).$$

The aggregated estimator is $\hat{V}^{s\to t}(\pi) = \frac{1}{M} \sum_{m=1}^{M} \hat{V}_m^{s\to t}(\pi)$ and $\hat{V}^t(\pi) = \sum_{s=1}^{S} w_s \hat{V}^{s\to t}(\pi)$.

Step 2: Decomposition Define the oracle estimator using true nuisance parameters:

$$V_m^{*s\to t}(\pi) = \frac{1}{n_{t,m}} \sum_{i \in \mathcal{D}_t^m} \mu_s^m(\pi(X_i), X_i) + \frac{n_s}{n_t} \frac{1}{n_{s,m}} \sum_{i \in \mathcal{D}_s^m} r_s^m(X_i) \frac{\mathbb{I}(A_i = \pi(X_i))}{e_s^m(A_i, X_i)} (Y_i - \mu_s^m(A_i, X_i))$$

and

$$V^{*s\to t}(\pi) := \frac{1}{M} \sum_{m=1}^{M} V_m^{*s\to t}(\pi).$$

The error decomposes as follows:

$$\begin{aligned} \sqrt{n_s}\big(\hat{V}^{s\to t}(\pi) - V^{s\to t}(\pi)\big) &= \underbrace{\sqrt{n_s}\big(\hat{V}^{s\to t}(\pi) - V^{*s\to t}(\pi)\big)}_{\text{Estimation error term}} + \underbrace{\sqrt{n_s}\big(V^{*s\to t}(\pi) - V^{s\to t}(\pi)\big)}_{\text{Oracle CLT term}} \\ &:= \Delta_1 + \Delta_2 \end{aligned}$$

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Step 3: Error Term Analysis We first analyze Δ_1:

$$
\begin{aligned}
&\hat{V}_m^{s\to t}(\pi) - V_m^{*s\to t}(\pi) \\
&= \frac{1}{n_{t,m}} \sum_{i\in\mathcal{D}_t^m} \hat{\mu}_s^m(\pi(X_i), X_i) + \frac{n_s}{n_t}\frac{1}{n_{s,m}} \sum_{i\in\mathcal{D}_s^m} \hat{r}_s^m(X_i)\frac{\mathbb{I}(A_i=\pi(X_i))}{\hat{e}_s^m(A_i,X_i)}(Y_i - \hat{\mu}_s^m(A_i,X_i)) \\
&- (\frac{1}{n_{t,m}} \sum_{i\in\mathcal{D}_t^m} \mu_s^m(\pi(X_i), X_i) + \frac{n_s}{n_t}\frac{1}{n_{s,m}} \sum_{i\in\mathcal{D}_s^m} r_s^m(X_i)\frac{\mathbb{I}(A_i=\pi(X_i))}{e_s^m(A_i,X_i)}(Y_i - \mu_s^m(A_i,X_i))) \\
&= \frac{1}{n_{t,m}} \sum_{i\in\mathcal{D}_t^m} \hat{\mu}_s^m(\pi(X_i), X_i) - \mu_s^m(\pi(X_i), X_i) \\
&+ \frac{n_s}{n_t}\frac{1}{n_{s,m}} \sum_{i\in\mathcal{D}_s^m} [\hat{r}_s^m(X_i)\frac{\mathbb{I}(A_i=\pi(X_i))}{\hat{e}_s^m(A_i,X_i)}(Y_i - \hat{\mu}_s^m(A_i,X_i)) - r_s^m(X_i)\frac{\mathbb{I}(A_i=\pi(X_i))}{e_s^m(A_i,X_i)}(Y_i - \mu_s^m(A_i,X_i))] \\
&:= \Delta_{11} + \Delta_{12}
\end{aligned}
$$

For each fold of source s, we then show $\mathbb{E}[\hat{V}_m^{s\to t}(\pi) - V_m^{*s\to t}(\pi) \mid \mathcal{D}^{-m}] = o_p(n_s^{-1/2})$.

Analysis of Δ_{11}:

$$
\mathbb{E}_{X\sim P_t}\big[\frac{1}{n_{t,m}} \sum_{i\in\mathcal{D}_t^m} \hat{\mu}_s^m(\pi(X_i), X_i) - \mu_s^m(\pi(X_i), X_i)\big] = \mathbb{E}_{X\sim P_t}\big[\hat{\mu}_s^m(\pi(X), X) - \mu_s(\pi(X), X)\big].
$$

Analysis of Δ_{12}: To simplify notations, here we use $e_\pi(X), e_A(X)$ to denote $e(\pi(X), X), e(A, X)$ respectively. The nuisance functions have been estimated by cross-fitting using the information from $\mathcal{D}^{-m}$.

$$
\begin{aligned}
&\mathbb{E}[\mathbb{I}(A=\pi(X_i))\left[\left(\frac{\hat{r}(X_i)}{\hat{e}_A(X_i)} - \frac{r(X_i)}{e_A(X_i)}\right)Y_i - \frac{\hat{r}(X_i)\hat{\mu}_A(X_i)}{\hat{e}_A(X_i)} + \frac{r(X_i)\mu_A(X_i)}{e_A(X_i)}\right]] \\
&= \mathbb{E}_X[e_\pi(X_i)\mu_\pi(X_i)\left(\frac{\hat{r}(X_i)}{\hat{e}_\pi(X_i)} - \frac{r(X_i)}{e_\pi(X_i)}\right) - \left(\frac{\hat{r}(X_i)\hat{\mu}_\pi(X_i)}{\hat{e}_\pi(X_i)} - \frac{r(X_i)\mu_\pi(X_i)}{e_\pi(X_i)}\right)] \\
&= \mathbb{E}_X[e_\pi(X_i)\mu_\pi(X_i)\frac{\hat{r}(X_i)}{\hat{e}_\pi(X_i)} - \frac{\hat{r}(X_i)\hat{\mu}_\pi(X_i)}{\hat{e}_\pi(X_i)}] \\
&= \mathbb{E}_X[\hat{r}(X_i)(\hat{\mu}_\pi(X_i) - \mu_\pi(X_i))\frac{e_\pi(X_i)}{\hat{e}_\pi(X_i)}]
\end{aligned}
$$

The second equality is from the iteration of expectation: $\mathbb{E}[\mathbb{I}(A=\pi(X))|X] = e(\pi(X), X)$ and $\mathbb{E}[\mathbb{I}(A=\pi(X))Y|X] = e(\pi(X), X)\mu(\pi(X), X)$. Note that the expectation is taken on the source data.

$$
\mathbb{E}_{X\sim P_s}\Big[\Delta_{12}\Big] = \frac{q_s}{q_t}\mathbb{E}_{X\sim P_s}\Big[\hat{r}^m(X)\frac{e_s(\pi(X),X)}{\hat{e}_s^m(\pi(X),X)}\big(\mu_s(\pi(X),X) - \hat{\mu}_s^m(\pi(X),X)\big)\Big]
$$

Combined Δ_{11} and Δ_{12}

$$
\begin{aligned}
&\mathbb{E}[\hat{V}_m^{s\to t} - V_m^{*s\to t} \mid \mathcal{D}^{-m}] \\
&= \mathbb{E}_{X\sim P_t}\big[\hat{\mu}_s^m(\pi(X),X) - \mu_s(\pi(X),X)\big] + \mathbb{E}_{X\sim P_s}\Big[\frac{q_s}{q_t}\hat{r}_s^m(X)\frac{e_s(\pi(X),X)}{\hat{e}_s^m(\pi(X),X)}\big(\mu_s(\pi(X),X) - \hat{\mu}_s^m(\pi(X),X)\big)\Big] \\
&= \mathbb{E}_{X\sim P_s}\Big[\frac{q_s}{q_t}r_s(X)\big(\hat{\mu}_s^m(\pi(X),X) - \mu_s(\pi(X),X)\big)\Big] \\
&+ \mathbb{E}_{X\sim P_s}\Big[\frac{q_s}{q_t}\hat{r}_s^m(X)\frac{e_s(\pi(X),X)}{\hat{e}_s^m(\pi(X),X)}\big(\mu_s(\pi(X),X) - \hat{\mu}_s^m(\pi(X),X)\big)\Big] \\
&= \frac{q_s}{q_t}\mathbb{E}_{P_s}\Big[\Big(\hat{r}_s^m(X)\frac{e_s}{\hat{e}_s^m} - r_s(X)\Big)(\mu_s - \hat{\mu}_s^m)\Big]
\end{aligned}
$$

The second equality is by importance weighting $P_t(X) = \frac{q_s}{q_t}r(X)P_s(X)$ ($P^t(X) := \mathbb{P}(x|G = t) = \frac{\mathbb{P}(G=t|x)p(x)}{\mathbb{P}(G=t)}$, $P^s(X) := \mathbb{P}(x|G=s) = \frac{\mathbb{P}(G=s|x)p(x)}{\mathbb{P}(G=s)}$)

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Since $\hat{e}_s^m \to e_s + o_p(1)$, the dominant term is:

$$\mathbb{E}_{P_s}\big[(\hat{r}_s^m(X) - r_s(X))(\mu_s(\pi(X), X) - \hat{\mu}_s^m(\pi(X), X))\big].$$

By Cauchy-Schwarz Inequality, we then bound this Second-Order Bias by Assumption 2.3:

$$\big|\mathbb{E}[\hat{V}_m^{s\to t} - V_m^{*s\to t} \mid \mathcal{D}^{-m}]\big| \le \|\hat{r}^m - r\|_{P_s}\, \|\hat{\mu}_s^m - \mu_s\|_{P_s} = o_p(n_s^{-1/2}).$$

Thus, the estimation error is $o_p(1)$ after $\sqrt{n_s}$-scaling.

Step 4: Asymptotic Distribution of the Oracle Term

Recall that the oracle estimator is

$$V^{*s\to t}(\pi) = \frac{1}{n_t}\sum_{i\in\mathcal{D}_t} \mu_s(\pi(X_i), X_i) + \frac{1}{n_t}\sum_{i\in\mathcal{D}_s} \frac{r_s(X_i)\,\mathbb{I}(A_i = \pi(X_i))}{e_s(A_i \mid X_i)}\big(Y_i - \mu_s(A_i, X_i)\big). \quad (22)$$

The population value is

$$V^{s\to t}(\pi) = \mathbb{E}_{X\sim P_t}\big[\mu_s(\pi(X), X)\big].$$

We then analyze Δ_2. Substituting Equation (22) yields

$$\Delta_2 = \sqrt{n_s}\Bigg[\frac{1}{n_t}\sum_{i\in\mathcal{D}_t}\Big(\mu_s(\pi(X_i), X_i) - V^{s\to t}(\pi)\Big) + \frac{n_s}{n_t}\cdot\frac{1}{n_s}\sum_{i\in\mathcal{D}_s}\frac{r_s(X_i)\,\mathbb{I}(A_i = \pi(X_i))}{e_s(A_i \mid X_i)}\big(Y_i - \mu_s(A_i, X_i)\big)\Bigg]. \quad (23)$$

We analyze the two terms separately.

The first term in Equation (23) can be rewritten as

$$\frac{\sqrt{n_s}}{\sqrt{n_t}}\cdot\frac{1}{\sqrt{n_t}}\sum_{i\in\mathcal{D}_t}\Big(\mu_s(\pi(X_i), X_i) - V^{s\to t}(\pi)\Big).$$

Since $\{X_i\}_{i\in\mathcal{D}_t}$ are i.i.d. from P_t and $\mathbb{E}_t[\mu_s(\pi(X), X)^2] < \infty$, the classical central limit theorem implies

$$\frac{1}{\sqrt{n_t}}\sum_{i\in\mathcal{D}_t}\Big(\mu_s(\pi(X_i), X_i) - V^{s\to t}(\pi)\Big) \xrightarrow{d} \mathcal{N}\big(0,\ \mathrm{Var}_t\big(\mu_s(\pi(X), X)\big)\big)\,.$$

By $n_s/n_t \to q_s/q_t$, Slutsky's theorem yields

$$\Delta_{21} \xrightarrow{d} \mathcal{N}\left(0,\ \frac{q_s}{q_t}\mathrm{Var}_t\big(\mu_s(\pi(X), X)\big)\right). \quad (24)$$

Define

$$\psi_s(Z) := \frac{r_s(X)\,\mathbb{I}(A = \pi(X))}{e_s(A \mid X)}\big(Y - \mu_s(A, X)\big).$$

Then the second term Δ_{22} in Equation (23) equals

$$\frac{n_s}{n_t}\frac{1}{\sqrt{n_s}}\sum_{i\in\mathcal{D}_s}\psi_s(Z_i).$$

By the law of iterated expectation,

$$\mathbb{E}_s[\psi_s(Z)] = \mathbb{E}_s\left[r_s(X)\mathbb{E}_s\left[\frac{\mathbb{I}(A = \pi(X))}{e_s(A \mid X)}\big(Y - \mu_s(A, X)\big)\ \Big|\ X\right]\right] = 0,$$

and under the overlap and boundedness assumptions, $\mathrm{Var}_s(\psi_s(Z)) < \infty$. Therefore,

$$\frac{1}{\sqrt{n_s}}\sum_{i\in\mathcal{D}_s}\psi_s(Z_i) \xrightarrow{d} \mathcal{N}(0,\ \mathrm{Var}_s(\psi_s(Z)))\,.$$

Using $n_s/n_t \to q_s/q_t$, Slutsky's theorem gives

$$\Delta_{22} \xrightarrow{d} \mathcal{N}\left(0,\ \frac{q_s^2}{q_t^2}\mathrm{Var}_s(\psi_s^m(Z))\right). \quad (25)$$

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Asymptotic distribution. Since the source and target samples are independent, the limits in Equation (24) and Equation (25) are independent. Combining them yields

$$\Delta_2 \xrightarrow{d} \mathcal{N}\big(0,\ \sigma^2_{s\to t}\big)\,,$$

where

$$\sigma^2_{s\to t} = \frac{q_s}{q_t}\mathrm{Var}_t\big(\mu_s(\pi(X), X)\big) + \frac{q_s^2}{q_t^2}\mathrm{Var}_s(\psi_s(Z)).$$

Finally, since the estimation error term $\Delta_1 = o_p(1)$, Slutsky's theorem implies

$$\sqrt{n_s}\big(\hat{V}^{s\to t}(\pi) - V^{s\to t}(\pi)\big) \xrightarrow{d} \mathcal{N}\big(0,\ \sigma^2_{s\to t}\big)$$

The final estimator is $\hat{V}^t(\pi) = \sum_{s=1}^S w_s \hat{V}^{s\to t}(\pi)$. We analyze the total error scaled by $\sqrt{n_S}$:

$$\begin{aligned}\sqrt{n}(\hat{V}^t(\pi) - V^t(\pi)) &= \sum_{s=1}^S w_s\sqrt{n}(\hat{V}^{s\to t}(\pi) - V^{s\to t}(\pi)) \\ &= \sum_{s=1}^S w_s \frac{1}{\sqrt{q_s}}\sqrt{n_s}(\hat{V}^{s\to t}(\pi) - V^{s\to t}(\pi)).\end{aligned}$$

Since the data generating process is i.i.d., the estimators from different sources are asymptotically independent. By Slutsky's theorem:

$$\sqrt{n}(\hat{V}^t(\pi) - V^t(\pi)) \xrightarrow{d} \mathcal{N}\left(0, \sigma^2_{total}\right), \tag{26}$$

where the total variance is:

$$\sigma^2_{total} = \sum_{s=1}^S \frac{w_s^2}{q_s}\sigma^2_{s\to t} = \sum_{s=1}^S w_s^2\Big(\frac{1}{q_t}\mathrm{Var}_t\big(\mu_s(\pi(X), X)\big) + \frac{q_s}{q_t^2}\mathrm{Var}_s(\psi_s(Z))\Big)$$

□

C Proof of Theorem 3.4 (Regret Analysis)

Proof. The multi-source regret $R(\pi)$ of a policy $\pi \in \Pi$ is defined in Equation (8) as:

$$R(\pi) := \max_{\pi'\in\Pi} Q(\pi') - Q(\pi).$$

We then decompose the regret and bound it.

$$\begin{aligned}R(\hat{\pi}) &= Q(\pi^*) - Q(\hat{\pi}) \\ &= Q(\pi^*) - \hat{Q}(\pi^*) + \hat{Q}(\pi^*) - \hat{Q}(\hat{\pi}) + \hat{Q}(\hat{\pi}) - Q(\hat{\pi}) \\ &\le Q(\pi^*) - \hat{Q}(\pi^*) + \hat{Q}(\hat{\pi}) - Q(\hat{\pi}) \\ &\le 2\sup_{\pi\in\Pi}|\hat{Q}(\pi) - Q(\pi)|\end{aligned}$$

Plugging in the definition of $Q(\pi)$ from Equation (2), we get

$$R(\hat{\pi}) \le 2\sup_{\pi\in\Pi}|\min_{s\in[S]}\{\hat{V}^{s\to t}(\pi)\} - \min_{s\in[S]}\{V^{s\to t}(\pi)\}|$$

By $|\min(a_i) - \min(b_i)| \le \max_i |a_i - b_i|$, we have

$$R(\hat{\pi}) \le 2\sup_{\pi\in\Pi}\max_{s\in[S]}|\hat{V}^{s\to t}(\pi) - V^{s\to t}(\pi)| \tag{27}$$

We first analyze $\sup_{\pi\in\Pi}|\hat{V}^{s\to t}(\pi) - V^{s\to t}(\pi)|$ for a single source s and finally take the maximum for all s. We decompose the error into the following two terms.

$$\begin{aligned}\sup_{\pi\in\Pi}|\hat{V}_{s\to t}(\pi) - V_{s\to t}(\pi)| &\le \sup_{\pi\in\Pi}|\hat{V}^{s\to t}(\pi) - V^{*s\to t}(\pi)| + \sup_{\pi\in\Pi}|V^{*s\to t}(\pi) - V^{s\to t}(\pi)| \\ &:= (i) + (ii).\end{aligned} \tag{28}$$

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where we define

$$V^{*s\to t}(\pi) := \frac{1}{n_t}\sum_{i\in\mathcal{D}_t}\mu_s(\pi(X_i),X_i) + \frac{1}{n_t}\sum_{i\in\mathcal{D}_s} r_s(X_i)\frac{\mathbb{I}(\pi(X_i)=A_i)(Y_i-\mu_s(A_i,X_i))}{e_s(A_i,X_i)}$$

Recall that

$$\hat{V}^{s\to t}(\pi) := \frac{1}{n_t}\sum_{i\in\mathcal{D}_t}\hat{\mu}_s(\pi(X_i),X_i) + \frac{1}{n_t}\sum_{i\in\mathcal{D}_s} \hat{r}(X_i)\frac{\mathbb{I}(A_i=\pi(X_i))}{\hat{e}_s(A_i,X_i)}\big(Y_i-\hat{\mu}_s(A_i,X_i)\big)$$

Bounding (i)

In this part, we show that the error from nuisance parameters is negligible to the order of $o_p(n_s^{-1/2})$. We split the data from source s into M folds. $\hat{\eta}_s = (\hat{r}_s, \hat{e}_s, \hat{\mu}_s)$ is estimated on the set $\mathcal{D}^{-m}$.

For simplicity, we define $\mathbb{I}_i(\pi) := \mathbb{I}(A_i = \pi(X_i))$. $\hat{\mu}_i := \hat{\mu}_s(A_i,X_i), \mu_{\pi,i} := \mu_s(\pi(X_i),X_i)$. $\psi_i(\hat{\eta}) := \hat{r}_i\frac{\mathbb{I}_i(\pi)(Y_i-\hat{\mu}_i)}{\hat{e}_i}$.

We decompose the error (i) into two parts.

$$\begin{aligned}
(i) &= \sup_{\pi\in\Pi}\left|\frac{1}{n_t}\sum_{i\in D_t}(\hat{\mu}_{\pi,i}-\mu_{\pi,i}) + \frac{1}{n_t}\sum_{i\in D_s}(\psi_i(\hat{r},\hat{e},\hat{\mu})-\psi_i(r,e,\mu))\right| \\
&= \sup_{\pi\in\Pi}\left|\frac{1}{n_t}\sum_{i\in D_t}(\hat{\mu}_{\pi,i}-\mu_{\pi,i}) + \frac{1}{n_t}\sum_{i\in D_s}(\psi_i(\hat{r},\hat{e},\hat{\mu})-\psi_i(r,e,\hat{\mu})) + \frac{1}{n_t}\sum_{i\in D_s}(\psi_i(r,e,\hat{\mu})-\psi_i(r,e,\mu))\right| \\
&\le \sup_{\pi\in\Pi}\left|\frac{1}{n_t}\sum_{i\in D_s}(\psi_i(\hat{r},\hat{e},\hat{\mu})-\psi_i(r,e,\hat{\mu}))\right| \\
&+ \sup_{\pi\in\Pi}\left|\frac{1}{n_t}\sum_{i\in D_t}(\hat{\mu}_{\pi,i}-\mu_{\pi,i}) + \frac{1}{n_t}\sum_{i\in D_s}(\psi_i(r,e,\hat{\mu})-\psi_i(r,e,\mu))\right|
\end{aligned}$$

Define term A

$$T_A := \sup_{\pi\in\Pi}\left|\frac{1}{n_t}\sum_{i\in D_s}(\psi_i(\hat{r},\hat{e},\hat{\mu})-\psi_i(r,e,\hat{\mu}))\right|$$

and term B

$$T_B := \sup_{\pi\in\Pi}\left|\frac{1}{n_t}\sum_{i\in D_t}(\hat{\mu}_{\pi,i}-\mu_{\pi,i}) + \frac{1}{n_t}\sum_{i\in D_s}(\psi_i(r,e,\hat{\mu})-\psi_i(r,e,\mu))\right|.$$

We first consider the term T_A.

$$\begin{aligned}
T_A &= \sup_{\pi\in\Pi}\left|\frac{1}{n_s}\sum_i\left(\frac{\hat{r}_i}{\hat{e}_i}-\frac{r_i}{e_i}\right)\mathbb{I}_i(\pi)(Y_i-\hat{\mu}_i)\right| \\
&\le \sup_{\pi\in\Pi}\left|\frac{1}{n_s}\sum_i\left(\frac{\hat{r}_i}{\hat{e}_i}-\frac{r_i}{e_i}\right)\mathbb{I}_i(\pi)(Y_i-\mu_i)\right| + \sup_{\pi\in\Pi}\left|\frac{1}{n_s}\sum_i\left(\frac{\hat{r}_i}{\hat{e}_i}-\frac{r_i}{e_i}\right)\mathbb{I}_i(\pi)(\mu_i-\hat{\mu}_i)\right| \\
&:= A_1 + A_2
\end{aligned}$$

We further bound the two terms separately.

Bounding A_1.

Define $h_1(Z_i;\pi(X_i)) := \left(\frac{\hat{r}_i}{\hat{e}_i}-\frac{r_i}{e_i}\right)\mathbb{I}_i(\pi)(Y_i-\mu_i)$. We have $A_1 = \sup_{\pi\in\Pi}\left|\frac{1}{n_s}\sum_i h_1(Z_i;\pi(X_i))\right|$.

We first show that $h_1(Z_i;\pi(X_i))$ is conditionally unbiased.

We divide the data from source $s\in[S]$ by M-fold, $\mathcal{D}^1,\dots,\mathcal{D}^M$. (For simplicity, we do not write s here.) For each fold m, the estimators of nuisance $(\hat{e}^m,\hat{\mu}^m,\hat{r}^m)$ are trained using the complementary

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data $\mathcal{D}^{-m} = \mathcal{D}/\mathcal{D}^m$. For any $i \in \mathcal{D}^m$, we have the following.

$$\begin{aligned}
\mathbb{E}[h_1(Z_i;\pi(X_i))] &= \mathbb{E}\left[\left(\frac{\hat{r}^m(X_i)}{\hat{e}_i^m(A_i|X_i)} - \frac{r^m(X_i)}{e_i^m(A_i|X_i)}\right)\mathbb{I}(A_i = \pi(X_i))(Y_i - \mu_i)\right] \\
&= \mathbb{E}\left[\left(\frac{\hat{r}^m(X_i)}{\hat{e}_i^m(\pi(X_i)|X_i)} - \frac{r^m(X_i)}{e_i^m(\pi(X_i)|X_i)}\right)\mathbb{I}(A_i = \pi(X_i))\mathbb{E}_{X,A}\left[(Y_i - \mu_s(\pi(X_i), X_i)\middle| X_i, A_i\right]\right] \\
&= 0
\end{aligned}$$

We then bound $h_1(Z_i;\pi)$ by the envelope $c_1(Z_i)$.

$$\begin{aligned}
\sup_{\pi\in\Pi}|h_1(Z_i;\pi)| &= \sup_{\pi\in\Pi}\left|\left(\frac{\hat{r}(X_i)}{\hat{e}(A_i|X_i)} - \frac{r(X_i)}{e(A_i|X_i)}\right)\mathbb{I}(A_i = \pi(X_i))(Y_i - \mu_i)\right| \\
&\le 2B_s \sup_{\pi\in\Pi}\left|\left(\frac{\hat{r}(X_i)}{\hat{e}(\pi(X_i)|X_i)} - \frac{r(X_i)}{e(\pi(X_i)|X_i)}\right)\right| := c_1(Z_i)
\end{aligned}$$

The inequality is from Assumption 2.1 $|Y_i - \mu_i| \le 2B_s$. We apply the bounded difference inequality conditional on X.

$$\begin{aligned}
&\mathbb{P}\left(\left|\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} h_1(Z_i,\pi(X_i))\right| - \mathbb{E}\left[\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} h_1(Z_i,\pi(X_i))\right||X\right]\right| \ge t\right) \\
&\le 2\exp(-\frac{2n_m^2t^2}{\sum_{i\in\mathcal{D}^m} c_1(Z_i)^2})
\end{aligned}$$

Take $t = \sqrt{\frac{\sum_{i\in\mathcal{D}^m} c_1(Z_i)^2\log(1/2\beta)}{2n_m^2}}$. With a probability of at least $1-\beta$, we have

$$\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} h_1(Z_i,\pi(X_i))\right| \le \mathbb{E}\left[\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} h_1(Z_i,\pi(X_i))\right||X\right] + \sqrt{\frac{\sum_{i\in\mathcal{D}^m} c_1(Z_i)^2\log(1/2\beta)}{2n_m^2}} \tag{29}$$

Let $Z_1', \ldots, Z_n'$ be an i.i.d. copy of $Z_1, \ldots, Z_n$, and $\epsilon_i \overset{i.i.d.}{\sim} \text{Unif}\{\pm 1\}$.

$$\begin{aligned}
&\mathbb{E}\left[\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} h_1\left(Z_i,\pi\left(X_i\right)\right)\right|\middle|X\right] \\
&= \mathbb{E}\left[\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} h_1\left(X_i,A_i,Y_i,\pi\left(X_i\right)\right) - \mathbb{E}_{A',Y'}\left[\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} h_1\left(X_i,A_i',Y_i',\pi\left(X_i\right)\right)\right]\right|\middle|X\right] \\
&= \mathbb{E}_Z\left[\sup_{\pi\in\Pi}\left|\mathbb{E}_{Z'}[\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} h_1\left(X_i,A_i,Y_i,\pi\left(X_i\right)\right) - \frac{1}{n_m}\sum_{i\in\mathcal{D}^m} h_1\left(X_i,A_i',Y_i',\pi\left(X_i\right)\right)|Z,X]\right|\middle|X\right] \\
&\le \mathbb{E}_Z\left[\mathbb{E}_{Z'}\left[\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} h_1\left(X_i,A_i,Y_i,\pi\left(X_i\right)\right) - \frac{1}{n_m}\sum_{i\in\mathcal{D}^m} h_1\left(X_i,A_i',Y_i',\pi\left(X_i\right)\right)\right|\middle|Z,X\right]\middle|X\right] \\
&= \mathbb{E}\left[\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} h_1\left(X_i,A_i,Y_i,\pi\left(X_i\right)\right) - \frac{1}{n_m}\sum_{i\in\mathcal{D}^m} h_1\left(X_i,A_i',Y_i',\pi\left(X_i\right)\right)\right|\middle|X\right] \\
&= \mathbb{E}\left[\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} \epsilon_i\left(h_1\left(X_i,A_i,Y_i,\pi\left(X_i\right)\right) - h_1\left(X_i,A_i',Y_i',\pi\left(X_i\right)\right)\right)\right|\middle|X\right] \\
&\le \mathbb{E}\left[\sup_{\pi\in\Pi}\left(\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} \epsilon_i h_1\left(X_i,A_i,Y_i,\pi\left(X_i\right)\right)\right| + \left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} \epsilon_i h_1\left(X_i,A_i,Y_i,\pi\left(X_i\right)\right)\right|\right)\middle|X\right] \\
&\le \mathbb{E}\left[\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} \epsilon_i h_1\left(X_i,A_i,Y_i,\pi\left(X_i\right)\right)\right|\middle|X\right] + \mathbb{E}\left[\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} \epsilon_i h_1\left(X_i,A_i',Y_i',\pi\left(X_i\right)\right)\right|\middle|X\right] \\
&= 2\mathbb{E}\left[\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} \epsilon_i h_1\left(X_i,A_i,Y_i,\pi\left(X_i\right)\right)\right|\middle|X\right]
\end{aligned}$$

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The first inequality above is due to: 1. Jensen's Inequality $|\mathbb{E}(x)| \leq \mathbb{E}(|x|)$. 2. The exchangeability of sup and $\mathbb{E}$. Therefore, we further derive Equation (29) as follows:

$$
\begin{aligned}
&\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} h_1\left(Z_i,\pi\left(X_i\right)\right)\right| \\
&\leq \mathbb{E}\left[\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} h_1(Z_i,\pi(X_i))\right||X\right] + \sqrt{\frac{\sum_{i\in\mathcal{D}^m} c_1(Z_i)^2\log(1/2\beta)}{2n_m^2}} \\
&\leq 2\mathbb{E}\left[\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m} \epsilon_i h_1\left(Z_i,\pi\left(X_i\right)\right)\right|\middle|X\right] + \sqrt{\frac{\sum_{i\in\mathcal{D}^m} c_1(Z_i)^2\log(1/2\beta)}{2n_m^2}} \\
&\leq \frac{\sqrt{\sum_{i\in\mathcal{D}^m} c_1(Z_i)^2}}{n_m}\cdot C\left(\kappa(\Pi)\right) + \sqrt{\frac{\sum_{i\in\mathcal{D}^m} c_1(Z_i)^2\log(1/2\beta)}{2n_m^2}} \\
&= \sqrt{\frac{4B_s^2\sum_{i\in\mathcal{D}^m}(|\hat{r}(X_i)/\hat{e}(X_i) - r(X_i)/e(X_i)|)^2}{n_m}}\cdot\frac{C\left(\kappa(\Pi)\right)+\sqrt{\log(1/2\beta)}}{\sqrt{n_m}} \\
&= o_p(\frac{2B_s(C\left(\kappa(\Pi)\right)+\sqrt{\log(1/2\beta)})}{\sqrt{n_m}})
\end{aligned}
\tag{30}
$$

The third inequality above is due to lemma A.4. The last equality is due to the following. Let $\delta(X) = r(X)/e(X)$. To bound $\|\hat{\delta} - \delta\|_n$, we use the decomposition $\frac{\hat{r}}{\hat{e}} - \frac{r}{e} = \frac{1}{\hat{e}}(\hat{r} - r) - \frac{r}{\hat{e}e}(\hat{e} - e)$. Under Assumption 2.1, the terms $1/\hat{e}$ and $r/(\hat{e}e)$ are bounded with probability approaching 1. By the triangle inequality and the consistency of the nuisance estimators:

$$\|\hat{\delta} - \delta\|_n \lesssim \|\hat{r} - r\|_n + \|\hat{e} - e\|_n = o_p(1)$$

For each fold, we have this convergence rate. Then, it also holds for the total data in source s because $n_m = n_s/M$.

$$
\begin{aligned}
A_1 &= \sup_{\pi\in\Pi}\left|\frac{1}{M}\sum_{m=1}^{M}\frac{1}{n_m}\sum_{i\in\mathcal{D}^m}(\hat{\delta}^m(X_i) - \delta(X_i))\mathbb{I}_i(Y_i - \mu_i)\right| \\
&\leq \frac{1}{M}\sum_{m=1}^{M}\sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m}(\hat{\delta}^m(X_i) - \delta(X_i))\mathbb{I}_i(Y_i - \mu_i)\right| \\
&= \frac{1}{M}\sum_{m=1}^{M} o_p(\frac{\sqrt{M}}{\sqrt{n_s}}) = o_p(\frac{1}{\sqrt{n_s}})
\end{aligned}
\tag{31}
$$

Bounding A_2. We derive the bound for the term A_2. Recall:

$$A_2 = \sup_{\pi\in\Pi}\left|\frac{1}{n_m}\sum_{i\in\mathcal{D}^m}(\hat{\delta}_i(\pi) - \delta_i(\pi))\mathbb{I}(A_i = \pi(X_i))(\mu_i - \hat{\mu}_i)\right|.$$

Applying the Cauchy-Schwarz Inequality to the empirical sum:

$$A_2 \leq \sup_{\pi\in\Pi}\underbrace{\left(\frac{1}{n_m}\sum_{i\in\mathcal{D}^m}(\hat{\delta}_i(\pi) - \delta_i(\pi))^2\mathbb{I}(A_i = \pi(X_i))\right)^{1/2}}_{\text{(I)}}\cdot\underbrace{\left(\frac{1}{n_m}\sum_{i\in\mathcal{D}^m}(\mu_i - \hat{\mu}_i)^2\right)^{1/2}}_{\text{(II)}}.$$

Conditional on the nuisance training set $\mathcal{D}^{-m}$, the estimators $\hat{\mu}, \hat{r}, \hat{e}$ are fixed.

First, consider Term (I). We evaluate the conditional expectation of the squared term with respect to the distribution P_s:

$$\mathbb{E}_{P_s}\left[(\hat{\delta}(\pi, X) - \delta(\pi, X))^2\mathbb{I}(A = \pi(X))\middle|\mathcal{D}^{-m}\right] \leq \|\hat{\delta}(\pi) - \delta(\pi)\|_{P_s,2}^2$$

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To bound $\|\hat{\delta}-\delta\|_{P_s,2}$, we use the decomposition $\frac{\hat{r}}{\hat{e}}-\frac{r}{e}=\frac{1}{\hat{e}}(\hat{r}-r)-\frac{r}{\hat{e}e}(\hat{e}-e)$. Under Assumption 2.1, the terms $1/\hat{e}$ and $r/(\hat{e}e)$ are bounded with probability approaching 1. By the triangle inequality:

$$\|\hat{\delta}-\delta\|_{P_s,2} \lesssim \|\hat{r}-r\|_{P_s,2}+\|\hat{e}-e\|_{P_s,2}.$$

By Markov's inequality, the empirical norm is of the same order as the L_2 norm. Combining the bounds for (I) and (II):

$$A_2 = O_p\left((\|\hat{r}-r\|_{P_s,2}+\|\hat{e}-e\|_{P_s,2})\cdot\|\hat{\mu}-\mu\|_{P_s,2}\right).$$

Under the rate assumption in Assumption 2.3, specifically that $\|\hat{\mu}-\mu\|_{P_s,2}\cdot\|\hat{r}-r\|_{P_s,2}=o_p(n_s^{-1/2})$ and $\|\hat{\mu}-\mu\|_{P_s,2}\cdot\|\hat{e}-e\|_{P_s,2}=o_p(n_s^{-1/2})$, we conclude:

$$A_2 = o_p(n_s^{-1/2}). \tag{32}$$

Bounding Term B

$$\begin{aligned}
|T_B| &= \sup_{\pi\in\Pi}\left|\frac{1}{n_t}\sum_{i\in D_t}(\hat{\mu}_{\pi,i}-\mu_{\pi,i})+\frac{1}{n_t}\sum_{i\in D_s}(\psi_i(r,e,\hat{\mu})-\psi_i(r,e,\mu))\right| \\
&= \sup_{\pi\in\Pi}\left|\frac{1}{n_t}\sum_{i\in\mathcal{D}_s}\left(r_i\frac{\mathbb{I}_i(\pi)(Y_i-\hat{\mu}_i)}{e_i}-r_i\frac{\mathbb{I}_i(\pi)(Y_i-\mu_i)}{e_i}\right)+\frac{1}{n_t}\sum_{i\in\mathcal{D}_t}(\hat{\mu}_{\pi,i}-\mu_{\pi,i})\right| \\
&= \sup_{\pi\in\Pi}\left|\frac{1}{n_s}\sum_{i\in\mathcal{D}_s}r_i\frac{\mathbb{I}_i(\pi)(\mu_i-\hat{\mu}_i)}{e_i}+\frac{1}{n_t}\sum_{i\in\mathcal{D}_t}(\hat{\mu}_{\pi,i}-\mu_{\pi,i})\right|
\end{aligned}$$

Write it using the full notation; the term T_B is defined as:

$$|T_B| = \sup_{\pi\in\Pi}\left|\frac{1}{n_t}\sum_{i\in\mathcal{D}_t}(\hat{\mu}(\pi(X_i),X_i)-\mu(\pi(X_i),X_i))+\frac{1}{n_t}\sum_{i\in\mathcal{D}_s}r_s(X_i)\frac{\mathbb{I}(A_i=\pi(X_i))}{e_s(A_i,X_i)}(\mu(A_i,X_i)-\hat{\mu}(A_i,X_i))\right|$$

For notation simplicity, let $\Delta_\mu(X,a):=\mu(a,X)-\hat{\mu}(a,X)$. Define $h_2(Z_i,\pi)$ as:

$$h_2(Z_i,\pi) := \frac{n}{n_t}\left[\mathbb{I}(G_i=t)\left(-\Delta_\mu(X_i,\pi(X_i))\right)+\mathbb{I}(G_i=s)\frac{r_s(X_i)}{e_s(A_i,X_i)}\mathbb{I}(A_i=\pi(X_i))\Delta_\mu(X_i,A_i)\right]$$

Thus, we rewrite T_B as the supremum of a sample mean:

$$|T_B| = \sup_{\pi\in\Pi}\left|\frac{1}{n}\sum_{i=1}^{n}h_2(Z_i,\pi)\right|$$

We then show that the expectation of $h_2(Z_i,\pi)$ given $\mathcal{D}^{-m}$ is zero. We first consider the second term inside the bracket. By the law of iterated expectations and $\mathbb{E}[\mathbb{I}(G=s)Z]=q_s\mathbb{E}[Z|G=s]$:

$$\begin{aligned}
&\mathbb{E}\left[\mathbb{I}(G=s)\frac{r_s(X)}{e_s(A,X)}\mathbb{I}(A=\pi(X))\Delta_\mu(X,A)\right] \\
&= q_s\mathbb{E}\left[\frac{r_s(X)}{e_s(A,X)}\mathbb{I}(A=\pi(X))\Delta_\mu(X,A)\middle|G=s\right] \\
&= q_s\mathbb{E}_{X|G=s}\left[\mathbb{E}\left[\frac{r_s(X)}{e_s(A,X)}\mathbb{I}(A=\pi(X))\Delta_\mu(X,A)\middle|X,G=s\right]\right] \\
&= q_s\mathbb{E}_{X|G=s}\left[r_s(X)\Delta_\mu(X,\pi(X))\right]
\end{aligned}$$

By the density ratio identity $r_s(x)=\frac{p(x|t)}{p(x|s)}\frac{q_t}{q_s}$:

$$\begin{aligned}
q_s\mathbb{E}_{X|G=s}\left[r_s(X)\Delta_\mu(X,\pi(X))\right] &= q_s\left(\frac{q_t}{q_s}\mathbb{E}_{X|G=t}[\Delta_\mu(X,\pi(X))]\right) \\
&= \mathbb{E}[\mathbb{I}(G=t)\Delta_\mu(X,\pi(X))]
\end{aligned}$$

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Summing the target and source components yields $\mathbb{E}[h_2|\mathcal{D}^{-m}] = 0$.

Based on the definition of h_2

1. If $i \in \mathcal{D}_t$:

$$|h_2(Z_i, \pi)| \leq \frac{n}{n_t} \max_{a \in \mathcal{A}} |\Delta_\mu(X_i, a)|$$

2. If $i \in \mathcal{D}_s$:

$$|h_2(Z_i, \pi)| = \left| \frac{n\mathbb{I}\{A_i = \pi(X_i)\}}{n_t} \cdot \frac{r_s(X_i)\Delta\mu(X_i, A_i)}{e_\pi(A_i|X_i)} \right| \leq \frac{nC_r}{n_t\eta_s} \max_{a \in \mathcal{A}} |\Delta_\mu(X_i, a)|$$

The inequality arises from the boundedness assumption Assumption 2.1. Thus, we define:

$$c_2(Z_i) := \frac{n}{n_t} \left(1 + \frac{C_r}{\eta_s}\right) \max_{a \in \mathcal{A}} |\mu(a, X_i) - \hat{\mu}(a, X_i)|$$

Therefore, we have the envelope function $c_2(Z_i)$ such that $|h_2(Z_i, \pi)| \leq c_2(Z_i)$ for all $\pi \in \Pi$.

Following a similar derivation of h_1, we bound the supremum of the empirical process using symmetrization with i.i.d. Rademacher variables ϵ_i.

$$\begin{aligned}
&\sup_{\pi \in \Pi} \left| \frac{1}{n_m} \sum_{i \in \mathcal{D}^m} h_2\left(Z_i, \pi\left(X_i\right)\right) \right| \\
&\leq 2\mathbb{E}\left[\sup_{\pi \in \Pi} \left| \frac{1}{n_m} \sum_{i \in \mathcal{D}^m} h_2(Z_i, \pi(X_i)) \right| |X \right] + \sqrt{\frac{\sum_{i \in \mathcal{D}^m} c_2(Z_i)^2 \log(1/2\beta)}{2n_m^2}} \\
&\leq 2\mathbb{E}\left[\sup_{\pi \in \Pi} \left| \frac{1}{n_m} \sum_{i \in \mathcal{D}^m} \epsilon_i h_2\left(Z_i, \pi\left(X_i\right)\right) \right| \middle| X \right] + \sqrt{\frac{\sum_{i \in \mathcal{D}^m} c_2(Z_i)^2 \log(1/2\beta)}{2n_m^2}} \\
&\leq \frac{\sqrt{\sum_{i \in \mathcal{D}^m} c_2(Z_i)^2}}{n_m} \cdot C\left(\kappa(\Pi)\right) + \sqrt{\frac{\sum_{i \in \mathcal{D}^m} c_2(Z_i)^2 \log(1/2\beta)}{2n_m^2}} \\
&\leq \sqrt{\frac{\sum_{i \in \mathcal{D}^m} \max_a |\Delta_\mu(X_i, a)|^2}{n_m}} \cdot \frac{C'\left(\kappa(\Pi)\right) + \sqrt{\log(1/2\beta)}}{\sqrt{n_m}} \\
&= O_P(\sum_a \|\hat{\mu}^m - \mu\|_{\mathbb{P}_s}) \cdot O_p(\frac{1}{\sqrt{n_m}}) \\
&= o_p(\frac{1}{\sqrt{n_s}})
\end{aligned} \tag{33}$$

Similarly, the third inequality is from lemma A.3. By Chebyshev's Inequality, we have the first equality above. The final equality is by Assumption 2.3. Combining Equation (31), Equation (32) and Equation (33), the total term (i) is asymptotically negligible.

Bounding (ii):

We analyze the stochastic term $\sup_{\pi \in \Pi} |V^{*s \to t}(\pi) - V^{s \to t}(\pi)|$ for a specific source s. Recall the structure of the oracle estimator $V^{*s \to t}(\pi)$ and the population value $V^{s \to t}(\pi) = \mathbb{E}_{P_t}[\mu_s(\pi(X), X)]$. By adding and subtracting the expectations of the component processes, we decompose the term into two independent empirical processes:

$$\begin{aligned}
&\sup_{\pi \in \Pi} \left| V^{*s \to t}(\pi) - V^{s \to t}(\pi) \right| \\
&= \sup_{\pi \in \Pi} \left| \frac{1}{n_t} \sum_{i \in \mathcal{D}_t} \mu_s(\pi(X_i), X_i) + \frac{n_s}{n_t} \frac{1}{n_s} \sum_{i \in \mathcal{D}_s} \psi_s(Z_i, \pi) - \mathbb{E}_{P_t}[\mu_s(\pi(X), X)] \right| \\
&\leq \underbrace{\sup_{\pi \in \Pi} \left| \frac{1}{n_t} \sum_{i \in \mathcal{D}_t} \mu_s(\pi(X_i), X_i) - \mathbb{E}_{P_t}[\mu_s(\pi(X), X)] \right|}_{(\mathcal{E}_t)} + \frac{n_s}{n_t} \underbrace{\sup_{\pi \in \Pi} \left| \frac{1}{n_s} \sum_{i \in \mathcal{D}_s} \psi_s(Z_i, \pi) \right|}_{(\mathcal{E}_s)}
\end{aligned}$$

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where $\psi_s(Z_i, \pi) := r_s(X_i)\frac{\mathbb{I}(A_i=\pi(X_i))}{e_s(A_i,X_i)}(Y_i - \mu_s(A_i, X_i))$. Note that $\mathbb{E}_{P_s}[\psi_s] = 0$. We now bound $\mathcal{E}_t$ and $\mathcal{E}_s$ separately with probability at least $1 - \beta/2$.

Analysis of $\mathcal{E}_t$:

Let $\mathcal{F}_t := \{x \to \mu_s(\pi(x), x) \mid \pi \in \Pi\}$ be the function class induced by policies in the target site.

By Assumption 2.1, potential outcomes are bounded by B_s. Thus, for any $f_\pi \in \mathcal{F}_t$, we have $|f_\pi(x)| \leq B_s$.

Applying lemma A.2, with probability of at least $1 - \beta/2$:

$$\mathcal{E}_t \leq 2\mathcal{R}_{n_t}(\mathcal{F}_t) + B_s\sqrt{\frac{2\log(4/\beta)}{n_t}}$$

We relate the covering number of $\mathcal{F}_t$ to the policy class Π. We consider $\pi_1, \pi_2 \in \Pi$. If two policies choose the same action $\pi_1(x) = \pi_2(x) = a$, then we have $\mu_s(\pi_1(x), x) - \mu_s(\pi_2(x), x) = 0$. Otherwise, $\pi_1(x) = a_1, \pi_2(x) = a_2, a_1 \neq a_2$. By the boundedness assumption, we have $|\mu_s(\pi_1(x), x) - \mu_s(\pi_2(x), x)| \leq |\mu_s(\pi_1(x), x)| + |\mu_s(\pi_2(x), x)| \leq 2B_s$ Therefore, the mapping $\pi \mapsto \mu_s(\pi(x), x)$ satisfies the Lipschitz condition:

$$|\mu_s(\pi_1(x), x) - \mu_s(\pi_2(x), x)| \leq 2B_s\mathbb{I}(\pi_1(x) \neq \pi_2(x))$$

This implies the covering number relationship $N(\epsilon, \mathcal{F}_t, \|\cdot\|_{n_t}) \leq N_H(\frac{\epsilon^2}{4B_s^2}, \Pi)$, where N_H is the Hamming covering number. Using Dudley's entropy integral (Lemma 5.2 in Appendix) with the substitution $u = \epsilon/(2B_s)$:

$$\mathcal{R}_{n_t}(\mathcal{F}_t) \leq \frac{24}{\sqrt{n_t}}\int_0^{B_s}\sqrt{\log N(\epsilon, \mathcal{F}_t)}d\epsilon \leq \frac{48B_s}{\sqrt{n_t}}\int_0^1\sqrt{\log N_H(u^2, \Pi)}du = \frac{48B_s\kappa(\Pi)}{\sqrt{n_t}}$$

Substitute the Rademacher bound back into the concentration inequality. With the probability of at least $1 - \beta/2$, we have:

$$\mathcal{E}_t \leq \frac{B_s}{\sqrt{n_t}}\left(96\kappa(\Pi) + \sqrt{2\log(4/\beta)}\right) \tag{34}$$

Analysis of $\mathcal{E}_s$: Let $\mathcal{F}_s := \{g_\pi(z) = \psi_s(z, \pi) \mid \pi \in \Pi\}$.

By the boundedness and overlap assumptions ($|Y - \mu_s| \leq 2B_s$, $e_s \geq \eta_s$) and the density ratio bound $|r_s| \leq C_r$:

$$|\psi_s(Z, \pi)| = \left|r_s(X)\frac{\mathbb{I}(A = \pi(X))}{e_s(A, X)}(Y - \mu_s(A, X))\right| \leq C_r\frac{1}{\eta_s}(2B_s)$$

Define $\Lambda_s := \frac{2C_rB_s}{\eta_s}$, which is the uniform upper bound for functions in $\mathcal{F}_s$.

By the concentration bound to the source samples (n_s), we have the following with the probability of at least $1 - \beta/2$:

$$\mathcal{E}_s \leq 2\mathcal{R}_{n_s}(\mathcal{F}_s) + \Lambda_s\sqrt{\frac{2\log(4/\beta)}{n_s}}$$

Using Dudley's Theorem (lemma A.3), the Rademacher complexity is bounded by the integral of the square root of the log covering number:

$$\mathcal{R}_{n_s}(\mathcal{F}_s) \leq \frac{24}{\sqrt{n_s}}\int_0^{\Lambda_s}\sqrt{\log N(\epsilon, \mathcal{F}_s, \|\cdot\|_{n_s})}d\epsilon \tag{35}$$

where Λ_s is the uniform envelope of the function class $\mathcal{F}_s$.

Similar to the target case, for any $\pi_1, \pi_2 \in \Pi$: $|\psi_s(Z, \pi_1) - \psi_s(Z, \pi_2)| \leq \Lambda_s|\mathbb{I}(A = \pi_1(X)) - \mathbb{I}(A = \pi_1(X))|$.

If two policies are the same, we have $|\mathbb{I}(A = \pi_1(X)) - \mathbb{I}(A = \pi_1(X))| = 0$. If $\pi_1(X) \neq \pi_2(X)$, we have $|\mathbb{I}(A = \pi_1(X)) - \mathbb{I}(A = \pi_1(X))| \leq 1$. This is because the indicator is less than 1 and the equality holds under $A = \pi_1(X)$ or $A = \pi_2(X)$. Therefore, we can bound the difference by $\mathbb{I}(\pi_1(X) \neq \pi_2(X))$:

$$|\mathbb{I}(A = \pi_1(X)) - \mathbb{I}(A = \pi_1(X))| \leq \mathbb{I}(\pi_1(X) \neq \pi_2(X)).$$

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In the end, we have

$$|\psi_s(Z,\pi_1) - \psi_s(Z,\pi_2)| \le \Lambda_s \mathbb{I}(\pi_1(X) \neq \pi_2(X)).$$

We can rewrite it as

$$\|\psi_s(\cdot,\pi_1) - \psi_s(\cdot,\pi_2)\|_{n_s} \le \Lambda_s \sqrt{d_H(\pi_1,\pi_2)}$$

where $d_H(\pi_1,\pi_2) = \frac{1}{n_s}\sum \mathbb{I}(\pi_1(X_i) \neq \pi_2(X_i))$ is the empirical Hamming distance. This inequality implies that an ϵ-cover of $\mathcal{F}_s$ in the L_2 norm can be constructed from a δ-cover of Π in the Hamming metric, provided that $\Lambda_s\sqrt{\delta} \le \epsilon$. Thus, we have the covering number relationship:

$$N(\epsilon, \mathcal{F}_s, \|\cdot\|_{n_s}) \le N_H\left((\epsilon/\Lambda_s)^2, \Pi\right) \tag{36}$$

Substituting Equation (36) into the integral Equation (35):

$$\begin{aligned}\int_0^{\Lambda_s} \sqrt{\log N(\epsilon, \mathcal{F}_s, \|\cdot\|_{n_s})}d\epsilon &\le \int_0^{\Lambda_s} \sqrt{\log N_H\left((\epsilon/\Lambda_s)^2, \Pi\right)}d\epsilon \\ &= \Lambda_s \int_0^1 \sqrt{\log N_H(u^2,\Pi)}du\end{aligned}$$

The equality above is from the change of variables $u = \frac{\epsilon}{\Lambda_s}$.

By definition, the Hamming entropy integral of the policy class is $\kappa(\Pi) := \int_0^1 \sqrt{\log N_H(u^2,\Pi)}du$. We obtain the final bound:

$$\mathcal{R}_{n_s}(\mathcal{F}_s) \le \frac{24\Lambda_s\kappa(\Pi)}{\sqrt{n_s}}$$

Substitute the Rademacher bound back into the concentration inequality. With a probability of at least $1-\beta/2$, we have:

$$\mathcal{E}_s \le \frac{\Lambda_s}{\sqrt{n_s}}\left(48\kappa(\Pi) + \sqrt{2\log(4/\beta)}\right) \tag{37}$$

where $\Lambda_s = \frac{2B_sC_r}{\eta_s}$.

Final Bound for The Stochastic Term We combine the bounds from Equation (34) and Equation (37). By the union bound, with probability at least $1-\beta$:

$$\begin{aligned}&\sup_{\pi\in\Pi}\left|V^{*s\to t}(\pi) - V^{s\to t}(\pi)\right| \\ &\le \frac{B_s}{\sqrt{n_t}}\left(96\kappa(\Pi) + \sqrt{2\log(4/\beta)}\right) + \frac{n_s}{n_t}\left[\frac{\Lambda_s}{\sqrt{n_s}}\left(48\kappa(\Pi) + \sqrt{2\log(4/\beta)}\right)\right]\end{aligned} \tag{38}$$

Final Aggregation

We bound the total regret of the learned policy $\hat{\pi}$ by the maximum estimation error across all source domains. Recall The regret Equation (27):

$$R(\hat{\pi}) \le 2\max_{s\in[S]}\sup_{\pi\in\Pi}|\hat{V}^{s\to t}(\pi) - V^{s\to t}(\pi)|.$$

Nuisance Bias: From the analysis of Term A and Term B, we have: $(i) = |T_A| + |T_B| = o_p(n^{-1/2})$ which is negligible compared with the stochastic term.

Stochastic Term: From the analysis of the oracle estimator, with probability of at least $1-\beta_2$ (plugging in $\Lambda_s = \frac{2B_sC_r}{\eta_s}$), we rewrite Equation (38)

$$(ii) \le \frac{B_s}{\sqrt{n_t}}(1 + \frac{2C_r}{\eta_s}\frac{\sqrt{n_s}}{\sqrt{n_t}})\left(96\kappa(\Pi) + \sqrt{2\log(4/\beta_2)}\right). \tag{39}$$

Final Regret Bound Applying a union bound over all S sources with $\beta_2 = \beta/S$, we obtain the final bound. With probability at least $1-\beta$:

$$R(\hat{\pi}) \le 2\max_{s\in[S]}\left[B_s(1 + \frac{2C_r}{\eta_s}\sqrt{\frac{n_s}{n_t}})\right]\frac{1}{\sqrt{n_t}}\left(96\kappa(\Pi) + \sqrt{2\log(4S/\beta)}\right) \tag{40}$$

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Under the asymptotic regime where $n_s \asymp n/S$ (balanced sources) and $n_t \asymp n_s$, the dominant term is of order $O(1/\sqrt{n/S}) = O(\sqrt{S/n})$. Thus:

$$R(\hat{\pi}) = O_p\left(\sqrt{\frac{S}{n}}\left(\kappa(\Pi) + \sqrt{\log S}\right)\right).$$

This confirms that the regret decays at the optimal parametric rate $\sqrt{n}$, modulated by the complexity of the policy class and the number of source domains. □

D Ablation Study Settings

For the source-number scaling ablation, we increase the number of source environments to $S \in \{4, 6, 8\}$ while keeping the covariate dimension fixed at 8 and the total number of source samples fixed at 9000. Thus, each source has 2250, 1500, and 1125 samples for $S = 4, 6, 8$, respectively. The source multiplier η_s is set to heterogeneous values, namely $(-8, -4, 3, 5)$ for $S = 4$, $(-8, -5, -2, 2, 4, 5)$ for $S = 6$, and $(-8, -5, -2, 1, 2, 3, 4, 5)$ for $S = 8$. Since visualizing the full high-dimensional simplex is infeasible, we evaluate policies on a dense grid over all pairwise simplex edges with a grid step 0.1, yielding 58, 141, and 260 evaluation weights for $S = 4, 6, 8$, respectively. As in the imbalance ablation, we report MS with $\tau = 0.5$ and use the same value and robust-regret summaries.

For the source sample imbalance ablation, we keep the same three-source data-generating process as in the main simplex-grid experiment, including the covariates, the same source-specific reward mechanisms, and the same simplex-grid evaluation protocol. The only change is the allocation of the fixed total source sample size across sources. Specifically, we consider imbalanced allocations such as $(1500, 3750, 3750)$, $(750, 4125, 4125)$, and $(4125, 4125, 750)$, while keeping the total number of source samples fixed at 9000. We report our method with $\tau = 0.5$ and evaluate the learned policies over the same simplex grid as in the main experiment.

E Experiments Compute Resources

We conduct all experiments on a computer equipped with an Intel Core i7-14700KF CPU and an NVIDIA GeForce RTX 4070 Ti SUPER GPU with 16 GB VRAM. Reproducing all the experimental results, including the baselines, takes a few hours.

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